College of Engineering
Chemical Engineering Department

King Saud University

College of Engineering

Chemical engineering Department

Graduation Project 2 CHE 496

Production of n-Butanol from Ethanol

BY

Mohammed Alrabiah 439100674

Yousef Rakan Freih 441106035

Supervised by

Dr. Khalid Alhumaizi

Dr. Abdelhamid Ajbar

1ST Semester

1447 H – 2025 G

Executive Summary

The production of n-butanol is undergoing a shift from fossil fuel-based methods to greener, renewable approaches. Traditionally derived from petrochemicals like propylene, n-butanol is now increasingly synthesized from ethanol using catalytic processes such as the Guerbet reaction. This method combines two ethanol molecules into n-butanol using Cu/ZrO_2. as a catalyst under controlled temperature and pressure, .offering better selectivity and energy efficiency

n-Butanol is a versatile chemical with wide industrial applications—as a solvent in paints and inks, an intermediate in producing plasticizers and resins, and as a potential biofuel due to its higher energy density compared to ethanol. Its demand is rising globally, particularly in the Asia-Pacific region, driven by growth in construction and .chemical manufacturing

In Saudi Arabia, the SABUCO facility in Jubail leads regional production, showcasing the Kingdom's strategy to diversify its chemical portfolio beyond oil. With sustainable feedstocks like bioethanol and efficient catalytic technology, the Guerbet process holds promises for environmentally friendly and economically viable .n-butanol production

Problem Statement

N-Butanol has been identified as a viable alternative to diesel and gasoline. It is regarded as a promising biofuel, capable of being utilized in vehicles designed for gasoline without necessitating engine modifications when used at 85 percent concentration, unlike 85% ethanol. Additionally, butanol offers a higher energy yield per volume compared to ethanol, approaching that of gasoline, which results in fuel consumption rates that are more aligned with gasoline than with ethanol. Furthermore, butanol can be blended with diesel to mitigate soot emissions. Historically, n-butanol has been synthesized from fossil fuels through methods such as the Reppe process, which utilizes propylene as a precursor. However, the growing emphasis on sustainable technologies has sparked interest in deriving n-butanol from ethanol, a renewable resource. In this project, students are tasked with developing a flowsheet for the catalytic production of 50,000 tons per year of n-butanol from ethanol. They will need to perform material and energy balances, design key equipment, conduct economic assessments, and pay particular attention to environmental impacts.

Table of Contents

Introduction 6

1.1 Background and History.....6

1.2 Applications and Uses.....7

1.3 Properties.....8

1.4 Initial feasibility study and market survey 10

2. **Summary of Mass and Energy Balance** 12

2.1. Summary of Mass and Energy Balance 12

2.2 Process flow sheet.....14

3. **introduction and objectives of equipment design** 15

3.1. introduction and objective of equipment design.....15

3.2. Equipment sizing 66

4. **Process control and instrumentation** 70

4.1. Types of controllers.....70

4.2 Various types of supervision Loops.....71

4.3. Types of feedback.....72

4.4. instrumentation.....72

5. **Hazard and environmental analysis**.....77

5.1. Hazop analsis.....77

5.2 Environmental Considerations.....81

5.3 inherent safety aspects and Emissions Table.....83

6. **Econmic analysis** 85

6.1. Capital cost Estimation.....85

6. Econmic analysis 90

7. **Contemporary issues**.....99

7.1. Energy integration 99

7.2. Process Optimization.....100

7.3. Water Treatment and Recycling.....100

7.4. Novel Techniques.....100

8. **Site Selection and Plant layout**.....102

8.1. Site Selection 102

8.2. Comparison of available indusrial cities.....107

8.3. Finle Site Selection.....108

8.4. Plant and equioment layout..... 109

9. **Conclusion and recommendation**.....115

9.1. Conclusion and recommendation 115

9.2. Reference.....117

List of figures

Figure 2.1 Flowsheet Selected Process Guerbet process. 14
Figure 3.1 Polymath Code for R-101……...21
Figure 3.2 Result for Reactor Volume…….22
Figure 3.3 Polymath result Report……23
Figure 3.4 Polymath equations for R-101……24
Figure 3.5 Volume of Reactor Vs Temperature inside reactor……25
Figure 3.6 Volume of Reactor Vs Conversion of Ethanol……..25
Figure 3.7 Volume of Reactor Vs Components Flow rates…….26
Figure 3.8 Correction factor chart……..33
Figure 3.9 j_h vs Reynold number chart for tube side……...34
Figure 3.10 j_f vs Reynold number chart for tube side……..35
Figure 3.11 Pitch arrangement types……36
Figure 3.12 Bundle clearance chart……..37
Figure 3.13 j_h vs Reynold number chart for shell side……38
Figure 3.14 j_f vs Reynold number chart for shell side 39
Figure 3.15 VLE data for H_2O- MA mixture system……...57
Figure 3.16 Selection of Centrifugal Pump…….67
Figure 3.17 Selection of Centrifugal Pump…….69
Figure 4.1 Control Scheme of Reactor……76
Figure 5.1 HAZOP Process Steps……78
Figure 6.1 Discounted cash flow diagram……...91
Figure 6.2 Non-discounted cash flow diagram……91
Figure 6.3 Discounted and non-discounted profitability Criterion……92
Figure 6.4 NPV curve data……..92
Figure 6.5 Discounted cash flow rate of return data curve……..93
Figure 6.6 DPBP curve data……94
Figure 6.7 Cumulative cash position data curve……..95
Figure 6.8 ROROI data curve……..96
Figure 6.9 Payback Period data curve……97
Figure 8.1 site selected for the Butanol Plant ……109
Figure 8.2 Equipment Layout of the Butanol Plant ……113

List of Tables

Table 1.1 physical properties 8
Table 1.2 Heat capacity.....9
Table 1.3 Viscosity.....9
Table 2.1 List of Assumptions 12
Table 2.2 Summary of material balance.....12
Table 2.3 Summary of energy balance.....13
Table 3.1 Stoichiometric table of Butanol Production Reaction……..16
Table 3.2 Specification Sheet of R-101……...28
Table 3.3 Heat Transfer coefficients……31
Table 3.4 Fouling factors……32
Table 3.5 Standard tube diameter values……33
Table 3.6 values of k and n……36
Table 3.7 Properties table……40
Table 3.8 H-104 Specification Sheet……45
Table 3.9 Specification Sheet for F-101……...46
Table 3.10 Material balance on the column results……49
Table 3.11 Antoine equation constants……51
Table 3.12 VLE data for Hexanol & Octanol system ……53
Table 3.13 Thickness Data……..56
Table 3.14 Specification Sheet of D-104……64
Table 3.15 guided words used in HAZOP analysis……64
Table 5.1 Representative HAZOP ……..77
Table 5.2 emission and allowable limit for the butanol production……80
Table 5.3 Equipment Cost……84
Table 6.1 working capital cost……86
Table 8.1 Site Comparison for Butanol Plant ……107
Table 8.2 Scoring of Final Site Options ……108

Chapter 1

Introduction

1.1 Background and History

The production of n-butanol from ethanol has evolved as a promising alternative to traditional petrochemical methods, driven by the need for more sustainable and renewable sources for chemical production. n-Butanol, a four-carbon alcohol, is a key raw material in the production of plastics, solvents, and other chemicals. Traditionally, n-butanol has been produced from fossil fuels using processes like the Reppe process, which relies on propylene as the starting material. However, the push for greener technologies has led to increasing interest in producing n-butanol from ethanol, a renewable feedstock.

The process of converting ethanol to n-butanol can be done via catalytic conversion or biological fermentation. The catalytic process involves a series of chemical reactions, One of the primary reactions is the hydrodeoxygenation of ethanol, where ethanol (C_2H_5OH) is converted to n-butanol (C_4H_9OH) by removing oxygen atoms.

Chemical Reaction:

$$\mathbf{2C2H5OH \rightarrow C4H9OH + H2O}$$

The Guerbet process is a well-established chemical reaction used for the production of higher alcohols, specifically butanol, from ethanol. This process involves the coupling of two molecules of ethanol to form butanol, typically under the influence of a metal catalyst. The Guerbet reaction is of significant interest due to its potential applications in the biofuel industry and the synthesis of industrial chemicals.

For the catalytic conversion of ethanol to butanol via the Guerbet reaction, an effective catalyst is Cu–Mg–Al mixed oxide (derived from hydrotalcite-type precursors) or Cu/ZrO_2. These catalysts are widely used industrially due to their bifunctional nature they provide both dehydrogenation and condensation sites essential for the Guerbet pathway. The copper component promotes ethanol dehydrogenation to acetaldehyde, while the basic sites on MgO or Al_2O_3 facilitate aldol condensation of intermediate species, leading to the formation of higher alcohols such as n-butanol. After condensation, the Cu sites also help in hydrogenation of crotonaldehyde intermediates, completing the reaction cycle.

The choice of catalyst is crucial because it determines the selectivity, conversion efficiency, and lifetime of the reactor. So, we select the Copper-based catalysts over nickel or cobalt ones because they minimize side reactions (like cracking or over-hydrogenation) and provide a cleaner product distribution with higher butanol yield. In addition, using Cu–Mg–Al oxides enhance thermal stability, resistance to sintering, and regeneration capability, making them ideal for long-term industrial operations.

1.2 Application and Uses

n-Butanol as a solvent

- Paints and Coatings: n-Butanol can be worked or used as a solvent in the formation of paints and coatings. Its ability to dissolve both synthetic and natural resins makes it ideal for use in varnishes, lacquers, and enamel paints. The solvent properties of n-Butanol help enhance the flow and spread paints, enhancing its completion and durability.

- Inks and Dyes: In the printing industry, N-butanol is valued for its effectiveness in adjusting the viscosity of inks. It helps in maintaining the consistency required for high-quality printing without affecting the color or drying time.

Intermediate in Chemical Synthesis

- Plasticizers: N-butanol serves as a key intermediate in the production of phthalate and non-phthalate plasticizers, which are used to enhance the flexibility, softness, and workability of plastics. These plasticizers are crucial in manufacturing a wide range of products, from flexible PVC tubing and cables to leather cloth and vinyl flooring.

- Butyl Acrylate: As a precursor to butyl acrylate, N-butanol is essential in the production of acrylic fibers, which are used in textiles, and resins, which are employed in adhesives, coatings, and sealants. Butyl acrylate imparts desirable properties such as improved weather resistance, low-temperature flexibility, and durability.

- Glycol Ethers: N-butanol is also used to produce various glycol ethers, which find applications in solvents for cleaning fluids, brake fluids, and as coupling agents in water-based coatings.

Biofuel Component

- Biofuel Production: N-butanol is being explored as a potential biofuel or biofuel additive. Due to its higher energy content compared to ethanol, N-butanol can be blended with gasoline to produce a fuel that is more efficient and less corrosive. Additionally, its ability to be blended at higher concentrations with gasoline without phase separation in the presence of water is a significant advantage.

Other Industrial Uses

- Extraction Agent: In pharmaceutical and chemical industries, N-butanol is used as an extraction solvent during the production processes of various drugs and fine chemicals. Its selectivity in solutions helps in isolating desired compounds effectively.

- Cosmetics and Flavors: N-butanol is occasionally used in the formulation of cosmetics and personal care products due to its solvent properties, which can stabilize volatile compounds like essential oils and fragrances.

1.3. Physical and Chemical Properties

Table 1.1 physical properties.

Component	Molecular formula	Mw (g/mol)	Tb (°C)	Tm(°C)	ρ (g/cm3)
Ethanol	C_2H_6O	46.069	78.5	-114	0.78945
Butanol	$C_4H_{10}O$	74.123	117.7	−89.8	0.81
Water	H_2O	18.01528	100	0	0.997
Hexanol	$C_6H_{14}O$	102.177	157	−45	0.82
Octanol	$C_8H_{18}O$	130.231	195	−16	0.83
Ethylene glycol	$C_2H_6O_2$	62.068	197.3	−12.9	1.1132
Hydrogen	H_2	1.008	−252.879	−259.16	0.08988

Table 1.2 Heat capacity.

Coefficients of C_p					
Components	A	B	C	D	E
Ethanol	59.342	3.64×10^{-1}	-1.22×10^{-3}	1.8×10^{-6}	0
H2	25.399	2.01×10^{-2}	-3.85×10^{-5}	3.188×10^{-8}	-8.76×10^{-12}
Butanol	83.877	5.66×10^{-1}	-1.72×10^{-3}	2.278×10^{-6}	0
Water	33.933	-8.40×10^{-3}	2.99×10^{-5}	-1.78×10^{-8}	3.69×10^{-12}
Hexanol	47.321	1.91	-4.83×10^{-3}	4.73×10^{-6}	0
Octanol	93.544	1.625	-4.11×10^{-3}	4.28×10^{-6}	0.00
Ethylene Glycol	75.88	0.64	-0.00164	0.00000169	0.00

Table 1.3 Viscosity.

Components	Molecular formula	Viscosity (μPa·s)
Ethanol	C_2H_6O	1.2
Butanol	$C_4H_{10}O$	2.573
Water	H_2O	0.890
Hexanol	$C_6H_{14}O$	4.58
Octanol	$C_8H_{18}O$	7.36
Ethylene Glycol	$C_2H_6O_2$	1.61×10^{-2}
Hydrogen	H2	8.76

1.4 Initial feasibility study and market survey

Any chemical production process relies on a sustainable flow of cash to exist. As chemical engineers, we must conduct a feasibility study to determine if a process is sustainable via the revenue it generates. Additionally, a market survey must collect feedback and gain insight into our target consumers to measure their inclination toward buying our products

Production and Consumption: Worldwide, Regional, and National

Butanol, especially n-butanol, plays a crucial role as an industrial solvent and chemical intermediate in various sectors, including paints, coatings, plastics, and biofuels. Its production and consumption patterns are influenced by regional industrial demand. Globally, in 2022, n-butanol consumption surpassed 5.3 million tons, with Asia Pacific leading, accounting for over 50% of the global demand, primarily driven by the rapid industrialization and urbanization of countries like China and India. North America and Western Europe together made up around 33% of global consumption, with the United States being a significant consumer due to its strong automotive, construction, and chemical industries. In Asia Pacific, China is the largest consumer, while India's market is growing rapidly. North America benefits from the shale gas boom, providing low-cost feed stocks for n-butanol production. Europe, particularly Germany, has a strong demand driven by its automotive and chemical sectors. The Middle East, with Saudi Arabia's SABUCO, and Africa, especially South Africa, are emerging players in production and export. Looking ahead, the global n-butanol market is projected to grow to 8.2 million tons by 2032, driven by increasing demand in construction, automotive, and chemical industries, with Asia Pacific, especially India, expected to lead this growth

Production of Butanol in Saudi Arabia.

Saudi Arabia is home to the world's largest butanol production facility, the Saudi Butanol Company (SABUCO), located in Jubail Industrial City. This joint venture comprises Sadara Chemical Company, Saudi Kayan Petrochemical Company, and Saudi Acrylic Acid Company (SAAC), with an investment of approximately $533 million.

SABUCO's plant boasts an annual production capacity of 330,000 tons of n-butanol and 11,000 tons of iso-butanol. These products are essential in manufacturing paints, coatings, adhesives, and various chemical derivatives. The facility utilizes propylene as its feedstock and is integrated with SAAC's acrylates complex, enhancing the .production of butyl acrylate—a key component in the coatings industry

The establishment of SABUCO underscores Saudi Arabia's strategic initiative to diversify its petrochemical portfolio and reduce reliance on traditional oil exports. By producing high-value chemical intermediates, the Kingdom aims to bolster its position in the global petrochemical market and support the growth of downstream industries.

Chapter 2

2.1 Summary of Mass and Energy Balance

Table 2.1 List of Assumptions

Process state	Steady state and ideal gas behavior
Plant operation	335 days
Shutdown period	35 days
Working period	24 hours/day
Production	50000 tons/year, butanol with 97% purity

Table 2.2 Summary of material balance

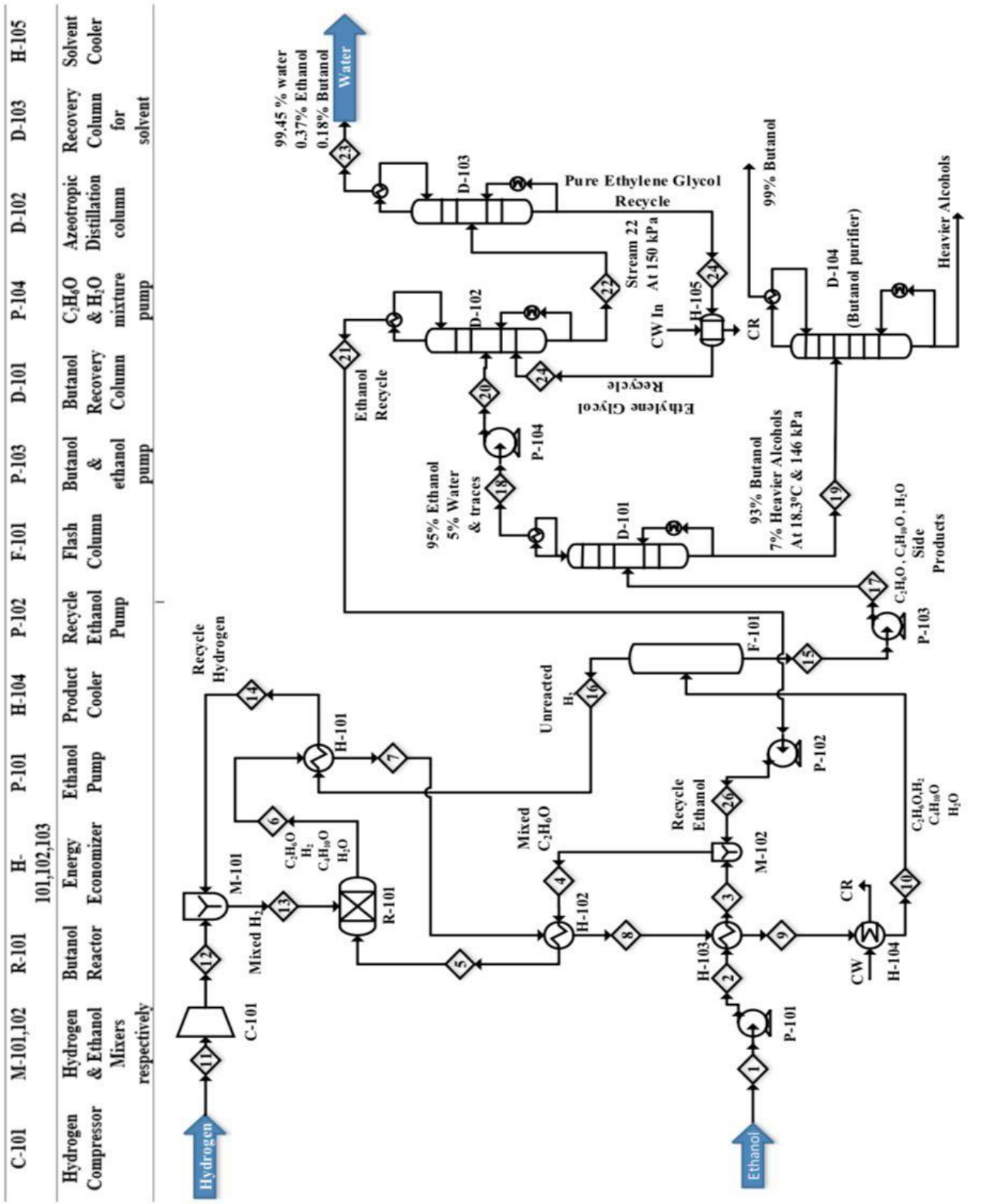
Unit	In(mol/s)	Out(mol/s)	
R-101	174.142 (Ethanol) 174.142 (Hydrogen)	6.6695 (Ethanol) 174.142 (Hydrogen) 77.037 (Butanol) 86.607 (Water) 1.914 (Hexanol) 1.914 (Octanol)	
F-101	6.6695 (Ethanol) 174.142 (Hydrogen) 77.037 (Butanol) 86.607 (Water) 1.914 (Hexanol) 1.914 (Octanol)	6.6695 (Ethanol) 77.037 (Butanol) 86.607 (Water) 1.914 (Hexanol) 1.914 (Octanol)	
D-101	6.6695 (Ethanol) 77.037 (Butanol) 86.607 (Water) 1.914 (Hexanol) 1.914 (Octanol)	77.037 (Butanol) 1.914 (Hexanol) 1.914 (Octanol)	
D-102	6.6695 (Ethanol) 250 (Ethylene Glycol) 86.607 (Water)	6.6695 (Ethanol Top Product) 250 (Ethylene Glycol Bottom product) 86.607 (Water)	
D-103	250 (Ethylene Glycol) 86.607 (Water)	86.607 (Water Top Product) 250 (Ethylene Glycol Bottom Product)	
D-104	77.037 (Butanol) 1.914 (Hexanol) 1.914 (Octanol)	77.037 (Butanol Top Product) 1.914 (Hexanol Bottom Product) 1.914 (Octanol Bottom Product)	

Table 2.3 Summary of energy balance

Unit	Energy (kW)
P-101	1.506
P-102	0.06
P-103	140
P-104	27
C-101	874
H-101	182
H-102	113
H-103	41.5
H-104	- 707
H-105	- 1708.71
R-101	-2829
D-101	$Q_c = -1020$ $Q_r = 1447.63$
D-102	$Q_c = -71.35$ $Q_r = 2417.48$
D-103	$Q_c = -951.3$ $Q_r = 3130.3$
D-104	$Q_c = -1362.2$ $Q_r = 1738.88$

2.2 Process flow sheet

Figure 2.1 Flowsheet Selected Process Guerbet process


C-101 Hydrogen Compressor
M-101,102 Hydrogen & Ethanol Mixers respectively
R-101 Butanol Reactor
H-101,102,103 Energy Economizer
P-101 Ethanol Pump
H-104 Product Cooler
P-102 Recycle Ethanol Pump
F-101 Flash Column
P-103 Butanol & ethanol pump
D-101 Butanol Recovery Column
P-104 C_2H_6O & H_2O mixture pump
D-102 Azeotropic Distillation column
D-103 Recovery Column for solvent
H-105 Solvent Cooler
Hydrogen
Ethanol
C-101
M-101
R-101
H-101
H-102
H-103
H-104
H-105
P-101
P-102
P-103
P-104
F-101
D-101
D-102
D-103
D-104
Mixed H_2
Recycle Hydrogen
Unreacted H_2
Mixed C_2H_6O
Recycle Ethanol
C_2H_6O , H_2 $C_4H_{10}O$ H_2O
C_2H_6O , $C_4H_{10}O$, H_2O Side Products
95% Ethanol 5% Water & traces
93% Butanol 7% Heavier Alcohols At 18.3°C & 146 kPa
Ethylene Glycol Recycle
Pure Ethylene Glycol Recycle
Stream 22 At 150 kPa
CW In
CR
CW
99.45 % water 0.37% Ethanol 0.18% Butanol
Water
99% Butanol
Heavier Alcohols
Ethanol Recycle

Chapter 3

3.1. Introduction and objectives of equipment design

Equipment design plays a vital role in ensuring efficient and safe production of butanol from ethanol through the Guerbet reaction. The process requires precise design of reactors, heat exchangers, and separation units to achieve high conversion and selectivity. Each equipment component must withstand high temperature and pressure while maintaining material integrity and corrosion resistance. Energy efficiency and heat recovery are key design objectives to minimize operational costs. Safety features such as pressure relief systems and automation controls ensure reliable operation. The reactor design must promote uniform mixing, proper catalyst contact, and effective heat removal. Economic feasibility is achieved through durable materials, low maintenance, and scalability. The design must also allow for future upgrades and process flexibility. Overall, well-engineered equipment ensures optimal performance and sustainability of the butanol production process.

3.2. Major Equipment Design

Butanol Production Reactor (R-101)

A Plug Flow Reactor (PFR), also known as a continuous tubular reactor, is characterized by the flow of reactants moving at a uniform velocity along the reactor's axis. In an ideal PFR, all substrate particles enter the reactor at one end, flow parallel to each other, and move through the reactor without any back-mixing. This flow pattern means that each element of fluid has a unique residence time based on its position in the reactor, which is directly proportional to the distance it has traveled along the length of the PFR

The PFR will be suitable for the Butanol production reaction. The inlet temperature of the reactants is 170ºC. As a result of exothermic reaction heat is produced inside the reactor. Process variable and design criteria:

Process variable:

- Rate of reaction $\frac{\text{moles}}{m^3 s}$ (r_A)

- Input volumetric flowrate $\frac{m^3}{s}$ (v_0)

- Volume of reactor m^3 (V)

Design criteria:

- Non adiabatic

- Non isothermal

- Conversion 96.17%

- Gas phase reaction

- Type of reactor: Plug Flow Reactor

- Material of construction: Stainless Steel

Detailed design of Butanol Production Reactor R-101

We set up a Stoichiometric Tables using Ethanol (C_2H_5OH) expressed as A as our basis of calculation in the following reaction, where we will use the stoichiometric tables to express the concentration as a function of flowrate.

$$\mathbf{2C_2H_5OH \rightarrow C_4H_9OH + H_2O}$$

$$2A \rightarrow C + D$$

$$\mathbf{14C_2H_5OH \rightarrow 2C_6H_{13}OH + 2C_8H_{17}OH + 10H_2O}$$

$$14A \rightarrow 2E + 2F + 10D$$

Table 3.1 Stoichiometric table of Butanol Production Reaction

Species	Feed flowrate (mol/s)	Change within reactor (mol/s)	Effluent rate from reactor (mol/s)
A	F_{A0}	$-F_{A0}X$	$F_A = F_{A0}(1-X)$
C	-	$F_{A0}X$	$F_C = F_{A0}X$
D	-	$11F_{A0}X$	$F_D = F_{A0}X$
E	-	$2F_{A0}X$	$F_E = F_{A0}X$
F	-	$2F_{A0}X$	$F_F = F_{A0}X$
Total	$F_{T0} = F_{A0}$		$F_T = F_A + F_C + F_D + F_E + F_F$

The differential equation developed for the reactor, relating conversion with the volume of reactor:

$$\frac{dX}{dV} = \frac{-r_A}{F_{A0}}$$

- X : Reaction conversion

- V: Volume of reactor in m^3

- r_A: Rate of reaction in moles per meter cube per unit time in seconds

- F_{A0} : Initial molar flowrate of Ethanol in mole per seconds

The rate of the reaction will be obtained from this equation:

$$-r_A = \frac{k_1 C_{Ethanol}}{1 + K_1 C_{Ethanol} + K_2 C_{Water}}$$

- $k_1 = 11.5\ \frac{1}{s}$:Rate constant in Arrhenius equation

- $K_1 = 1.9 \times 10^{-3} \frac{m^3}{mol}$

- $K_2 = 0.4 \times 10^{-3} \frac{m^3}{mol}$

Calculate the partial pressures from Gas equation as shown below:

$$PV = nRT$$

Therefore, $P_A = C_A RT$, $P_F = C_F RT$, $P_E = C_E RT$, $P_C = C_C RT$, and $P_D = C_D RT$

Where the concentration of each component in the reaction was calculated using the following equation as well:

$$C_i = \frac{F_i}{v}$$

Therefore, $C_A = \frac{F_{A0}(1-X)}{v}$, $C_C = \frac{F_{A0}X}{v}$, and $C_D = \frac{11F_{A0}X}{v}$, $C_E = \frac{2F_{AO}X}{v}$, $C_F = \frac{2F_{AO}X}{v}$

, and C_D : Molar Concentration of components in the reaction in C_A, C_E, C_F, C_C mole per meter cube

The volumetric flow rate was calculated by the equation shown below:

$$v = v_0(1 + \varepsilon X)\frac{T}{T_0}\frac{P_0}{P}$$

- v :Volumetric flow rate

- v_0 :Initial volumetric flowrate in meter cube per seconds

- ε : Relationship between Net mole change for the reaction and initial mole fraction of A(Ethanol)

- P_0 :Iinitial pressure in Bar

- P: Pressure in Bar

- T_0-Initial tempreture in kelvin

- T: Temperature in kelvin

Calculate the initial volumetric flow rate:

$$C_{A0} = \frac{F_{A0}}{v_0} \rightarrow C_{A0} = \frac{P_{A0}}{RT_0} \quad = \frac{(5 \times 0.5)\text{bar}}{8.314 \times 10^{-5}\text{bar} \cdot \frac{\text{m}^3}{\text{mol. K}} \times 443\ \text{K}} = 67.87 \frac{\text{mol}}{\text{m}^3}$$

So, we can now calculate

$$v_0 = \frac{F_{A0}}{C_{A0}} = \frac{48.3 \frac{\text{mol}}{\text{s}}}{67.87 \frac{\text{mol}}{\text{m}^3}} = 0.7116 \frac{\text{m}^3}{\text{s}}$$

Net moles change in total number of moles per mole of A reacted:

$$\sigma = \frac{\text{d}}{a} + \frac{\text{c}}{a} - \frac{a}{a} = \frac{1}{2} + \frac{1}{2} - 1 = 0$$

For second reaction

$$\sigma = \frac{\text{e}}{a} + \frac{\text{f}}{a} + \frac{d}{a} - \frac{a}{a} = \frac{2}{14} + \frac{2}{14} + \frac{10}{14} - 1 = 0$$

Relationship between Net mole change for the reaction and initial mole fraction of A (Butane):

$$y_{A0} = \frac{F_{A0}}{F_{A0} + F_{Hydrogen}} = 0.5$$
$$\varepsilon = y_{A0} \times \sigma = 0.5 \times 0 = 0$$

- y_{A0} : initial mole fraction of A (Ethanol)

- σ : Net moles change in total number of moles per mole of A reacted

The differential equation developed for the reactor, relating temperature with the volume of reactor:

$$\frac{dT}{dV} = \frac{Q + r_A \Delta H_{r \times n}(T)}{F_{AO}(\Sigma \theta_i C_{Pi} + \Delta C_P X)}$$

- T:Tempreture in Celsius

- W: Weight of catalyst in kg

- Q: Heat required for the reaction in joule per second

- r_A : Rate of reaction in moles per meter cube per unit time in seconds

- $\Delta H_{rxn}(T)$: Enthalpy of the reaction in joule per mole

- C_P : Specific heat in joule per mole per Celsius

Note: Values of Q, ΔC_P and C_P were already calculated in energy balance

$$\sum \theta_i C_{Pi} = \theta_A C_{PA} + \theta_F C_{PF} + \theta_E C_{PE} + \theta_C C_{PC} + \theta_D C_{PD} = 1 \times 87.5$$

$$\sum \theta_i C_{Pi} = 87.5 \frac{J}{\text{mol}^\circ\text{C}}$$

Therefore,

$$\frac{dT}{dV} = \frac{\left(-2829 \times 10^3 \frac{J}{\text{s}}\right) + \text{r}_\text{A} \times -670000 \frac{\text{J}}{\text{mol}}}{48.3 \frac{\text{mol}}{\text{s}}\left(87.5 \frac{\text{J}}{\text{mol.}^\circ\text{C}} + 28 \frac{\text{J}}{\text{mol.}^\circ\text{C}} \times \text{X}\right)} \times \frac{1}{1000}$$

Polymath Code

```
# Differential Equations & Initial Conditions
d(X)/d(V) = -rA / FA0
X(0) = 0
d(T)/d(V) =(((2829000 + rA*(-22668000))) / (48.3 * (87.5 +( 28* X)))) *0.0001
T(0) = 443
# Rate Equation and concentrations
rA = -(k1 * CA)/(1+(K1*(CA))+(K2*(CC)))
k1=11.5
K1=1.9
K2=0.4
CA = (FA0* (1 - X)) / v
CC = (FA0* X) / v
CD= (FA0*11*X) /v
CE= (FA0*2*X) /v
CF= (FA0*2*X) /v

v = (1 + ep * X) * (T / T0) * (P / P0) * v0
v0 = 0.7116
ep = yA0 * 0

# Initial conditions
P0 = 5
T0=443
FB0 = 48.3
FA0 = 48.3
yA0 = FA0 / (FA0 + FB0)

V(0) = 0
V(f) = 25
P=4.995
Yc=Fc/FA0
Fc=0.455*FA0*X
FD=0.513*FA0*X
FA=FA0*(1-X)
FE=0.011*FA0*X
FF=0.011*FA0*X
```

Ln 8 | Butanol Reactor.pol | No Title

Figure 3.1 Polymath Code for R-101

Results of the Polymath

Since in material balance we have considered the conversion of 96% of Ethanol to Butanol so the volume required for this conversion is

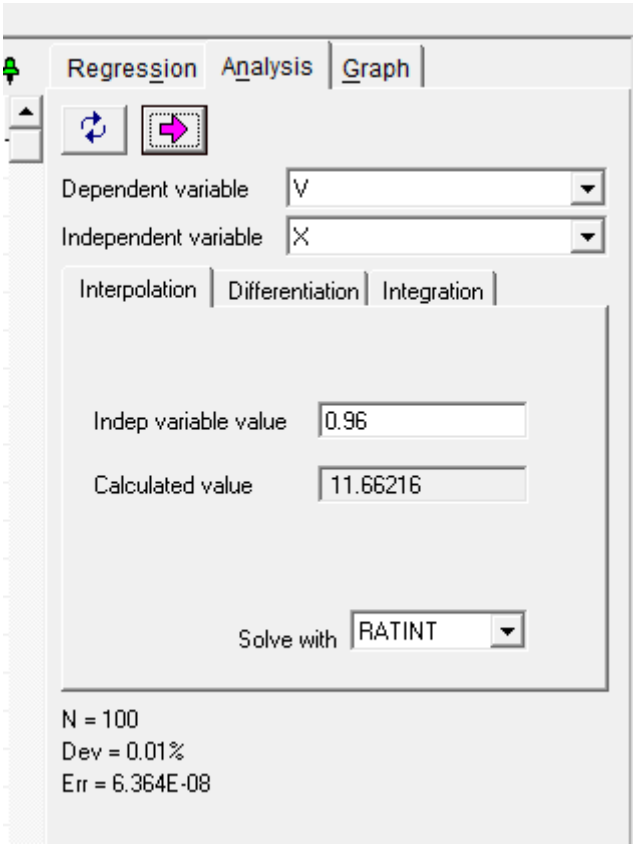


Figure 3.2 Result for Reactor Volume

Report of Polymath code

Calculated values of DEQ variables

	Variable	Initial value	Minimal value	Maximal value	Final value
1	CA	67.94315	0.0014282	67.94315	0.0014282
2	CC	0	0	64.49148	64.4756
3	CD	0	0	709.4062	709.2316
4	CE	0	0	128.983	128.9512
5	CF	0	0	128.983	128.9512
6	ep	0	0	0	0
7	FA	48.3	0.0010699	48.3	0.0010699
8	FA0	48.3	48.3	48.3	48.3
9	FB0	48.3	48.3	48.3	48.3
10	Fc	0	0	21.97601	21.97601
11	FD	0	0	24.77735	24.77735
12	FE	0	0	0.5312882	0.5312882
13	FF	0	0	0.5312882	0.5312882
14	k1	11.5	11.5	11.5	11.5
15	K1	1.9	1.9	1.9	1.9
16	K2	0.4	0.4	0.4	0.4
17	P	4.995	4.995	4.995	4.995
18	P0	5.	5.	5.	5.
19	rA	-6.006106	-6.006106	-0.000613	-0.000613
20	T	443.	443.	466.8146	466.8146
21	T0	443.	443.	443.	443.
22	v	0.7108884	0.7108884	0.749104	0.749104
23	V	0	0	25.	25.
24	v0	0.7116	0.7116	0.7116	0.7116
25	X	0	0	0.9999778	0.9999778
26	yA0	0.5	0.5	0.5	0.5
27	Yc	0	0	0.4549899	0.4549899

No File | POLYMATH Report

Figure 3.3 Polymath result Report

Explicit equations

```
1  FA0 = 48.3
2  k1 = 11.5
3  K1 = 1.9
4  K2 = 0.4
5  P0 = 5
6  T0 = 443
7  FB0 = 48.3
8  yA0 = FA0 / (FA0 + FB0)
9  ep = yA0 * 0
10 P = 4.995
11 v0 = 0.7116
12 v = (1 + ep * X) * (T / T0) * (P / P0) * v0
13 CC = (FA0* X) / v
14 CA = (FA0* (1 - X)) / v
15 rA = -(k1 * CA)/(1+(K1*(CA))+(K2*(CC)))
16 CD = (FA0*11*X) /v
17 CE = (FA0*2*X) /v
18 CF = (FA0*2*X) /v
19 Fc = 0.455*FA0*X
20 Yc = Fc/FA0
21 FD = 0.513*FA0*X
22 FA = FA0*(1-X)
23 FE = 0.011*FA0*X
24 FF = 0.011*FA0*X
```

Figure 3.4 Polymath equations for R-101

Graphs from polymath

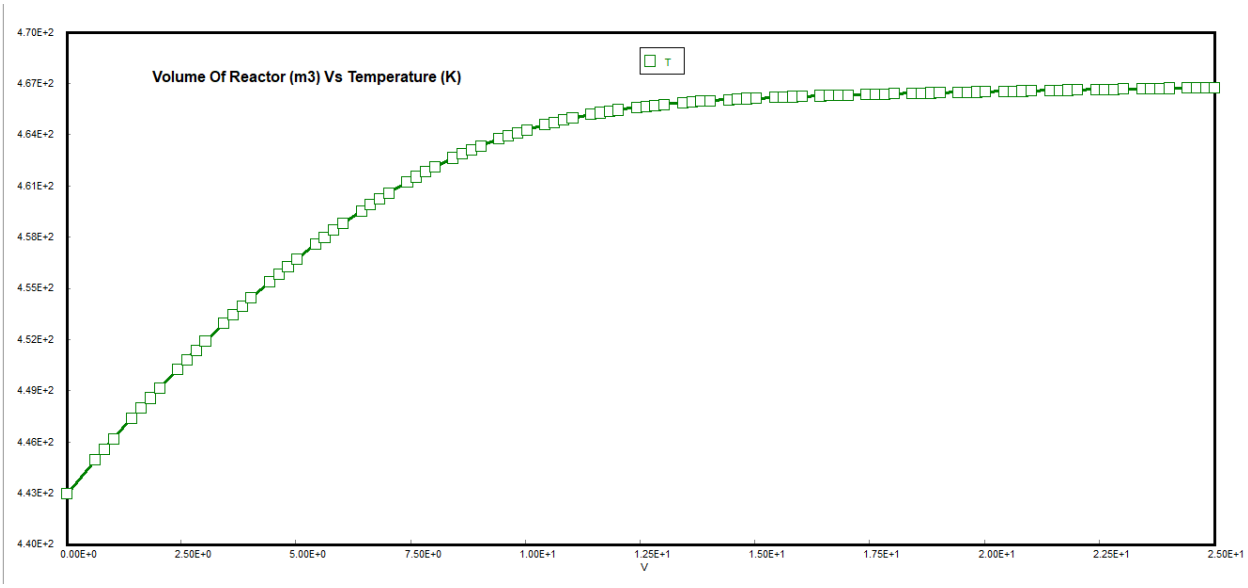


Figure 3.5 Volume of Reactor Vs Temperature inside reactor

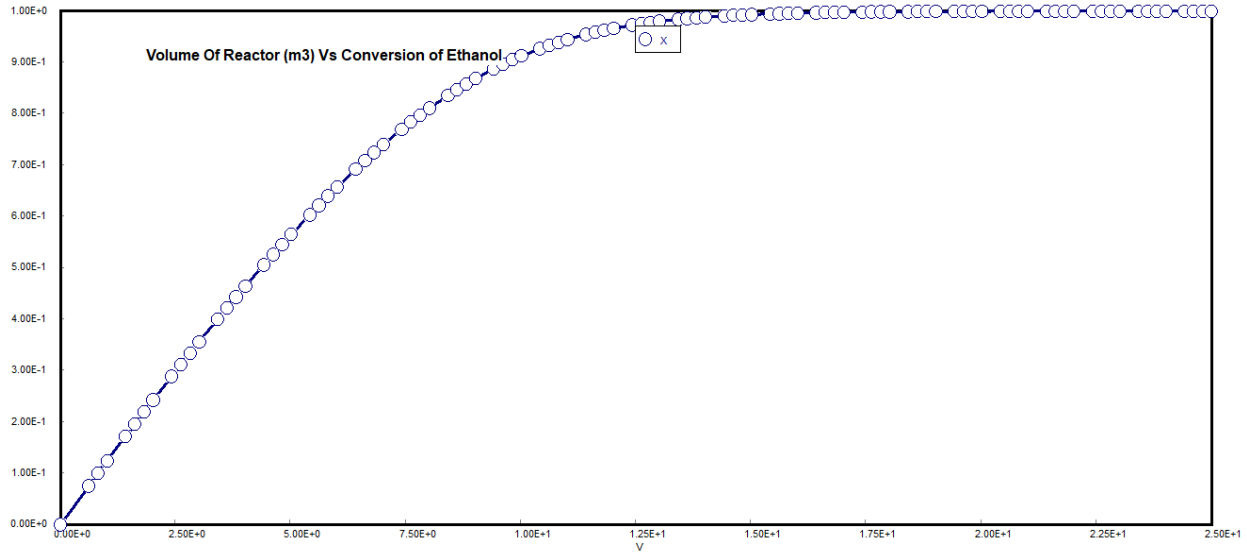


Figure 3.6 Volume of Reactor Vs Conversion of Ethanol

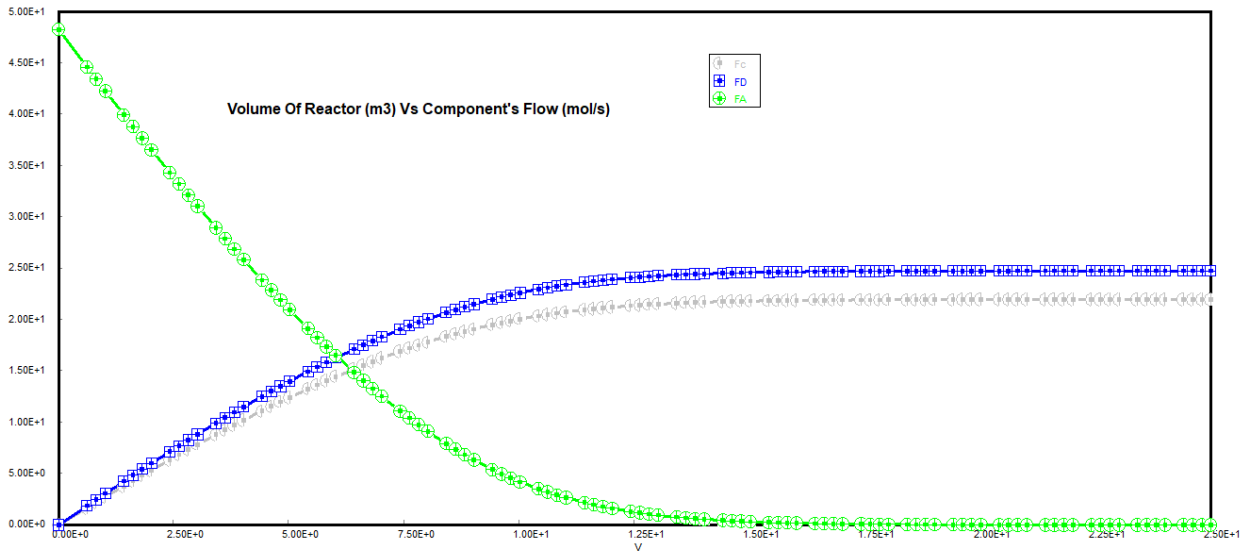


Figure 3.7 Volume of Reactor Vs Components Flow rates

Volume of Reactor

For the 96% conversion of Ethanol to Butanol, Hexanol and Octanol the reactor volume required is 11.66 m^3. For side reactions and safety margin for our system 10% overdesign of reactor.

$$Volume\ of\ reactor = 11.66 + 0.1 \times 11.66 = 12.83\ m^3$$

Residence Time of Reactor

The residence time of PFR is given by

$$\tau = \frac{V}{v_0} = \frac{12.83\ m^3}{0.7116\ m^3/s} = 13.298\ s$$

Height & Diameter of the Reactor

From rule of thumbs, PFR $\frac{H}{D} = 3\ to\ 5$.

So, we select $\frac{H}{D} = 3\ and\ H = 3D$

$$Volume\ of\ reactor = \frac{\pi D^2 H}{4}$$

Put $H = 3D$ in the above equation

$$Volume\ of\ reactor = \frac{3\pi D^3}{4}$$

$$D = \sqrt[3]{\frac{12.83 \times 4}{3\pi}} = 1.759\ m$$

Since $\frac{H}{D} = 3\ and\ H = 3D$ so the height of the reactor is

$$H = 3D = 3 \times 1.795 = 5.27\ m$$

Catalyst used in Reactor

For the catalytic conversion of ethanol to butanol via the Guerbet reaction, an effective catalyst is Cu–Mg–Al mixed oxide (derived from hydrotalcite-type precursors) or Cu/ZrO_2. These catalysts are widely used industrially due to their bifunctional nature they provide both dehydrogenation and condensation sites essential for the Guerbet pathway. The copper component promotes ethanol dehydrogenation to acetaldehyde, while the basic sites on MgO or Al_2O_3 facilitate aldol condensation of intermediate species, leading to the formation of higher alcohols such as n-butanol. After condensation, the Cu sites also help in hydrogenation of crotonaldehyde intermediates, completing the reaction cycle.

The choice of catalyst is crucial because it determines the selectivity, conversion efficiency, and lifetime of the reactor. So we select the Copper-based catalysts over nickel or cobalt ones because they minimize side reactions (like cracking or over-hydrogenation) and provide a cleaner product distribution with higher butanol yield. In addition, using Cu–Mg–Al oxides enhances thermal stability, resistance to sintering, and regeneration capability, making them ideal for long-term industrial operations.

Table 3.2 Specification Sheet of R-101

Parameter	Value
Molar flow rate of Ethanol	174 kmol/hr
Conversion of Ethanol	96%
Volume of Reactor	$m^3$12.83
Diameter of Reactor	1.76 m
Height of Reactor	5.27 m
Residence Time	13.29 s

3.2.2 Heat Exchanger (H-104)

Introduction

A heat exchanger is a device that transfers thermal energy (enthalpy) between two or more fluids, a fluid and a solid surface, or solid particles and a fluid at different temperatures while in thermal contact.

Heat Exchanger Design

Plant streams in most industrial processes need to be heated or cooled. The most efficient and long-standing approach in the industry is to use heat exchangers. A heat exchanger transfers heat between two fluids of differing temperatures without mixing them. There are several types of heat exchangers available, including double pipe, shell-and-tube, coil-and-shell, plate, and others. Each kind has benefits and limitations and is best suited for a certain purpose. Shell-and-tube heat exchangers (with one shell pass and two tube passes) are often utilized for best performance because they have a large surface area and a good dimensional ratio.

Assumptions and basis of calculation

- The heat exchangers used are shell-and-tube, with two tube passes and one shell pass.
- The design uses a split-ring floating head layout.
- The heat exchanger tubes form a triangle design.
- A 25% baffle cut was used for best performance.
- Performance parameters were found by trial-and-error.

Design Procedure

The main steps of design as follows:

- Use an energy balance to calculate heat exchanger duty.
- Determine the qualities of hot and cold fluids.
- Determine the Log Mean Temperature Difference (LMTD) and the actual temperature difference.
- Assume an overall heat transfer coefficient (U_o) and calculate the provisional heat transfer area.
- Determine the configuration of the tube and shell sides.
- Determine heat transfer area.

- Choose the arrangement and size of the tubes.
- Determine the number of tubes necessary.
- Determine the bundle and shell diameters.
- Calculate mass velocity and Reynolds number.
- Determine heat transfer coefficients for the inner (hi) and outer (ho) sides.
- Recalculate the total heat transfer coefficient (U_o) to confirm the first assumption.
- Determine the pressure drop across the heat exchanger.

Detailed Steps

- Specify the target fluid's intake and output temperatures, together with its mass flow rate.
- Use the sensible heat change formula to compute the amount of heat lost or supplied based on other fluid conditions.

$$\mathbf{Q = \dot{m} C_p \Delta T}$$

- Assume an overall heat transfer coefficient (U) based on Table-1 or other reliable references.
- Identify and define the type of shell-and-tube heat exchanger to employ.
Determine fouling variables on both tube and shell sides from Table.
- Assign fluids to the tube or shell side, considering factors such as fouling tendency, pressure, corrosiveness, and other safety considerations.

Table 1.3 Heat Transfer coefficients

Shell and Tube Exchangers		
Hot Fluid	**Cold Fluid**	***U* (W/m²°C)**
Heat exchangers		
Water	Water	800–1500
Organic solvents	Organic solvents	100–300
Light oils	Light oils	100–400
Heavy oils	Heavy oils	50–300
Gases	Gases	10–50
Coolers		
Organic solvents	Water	250–750
Light oils	Water	350–900
Heavy oils	Water	60–300
Gases	Water	20–300
Organic solvents	Brine	150–500
Water	Brine	600–1200
Gases	Brine	15–250
Heaters		
Steam	Water	1500–4000
Steam	Organic solvents	500–1000
Steam	Light oils	300–900
Steam	Heavy oils	60–450
Steam	Gases	30–300
Dowtherm	Heavy oils	50–300
Dowtherm	Gases	20–200
Flue gases	Steam	30–100
Flue	Hydrocarbon vapors	30–100
Condensers		
Aqueous vapors	Water	1000–1500
Organic vapors	Water	700–1000
Organics (some noncondensables)	Water	500–700
Vacuum condensers	Water	200–500
Vaporizers		
Steam	Aqueous solutions	1000–1500
Steam	Light organics	900–1200
Steam	Heavy organics	600–900

Table 3.4 Fouling factors

Fluid	Coefficient ($W/m^2\,°C$)	Factor (resistance) ($m^2\,°C/W$)
River water	3000–12,000	0.0003–0.0001
Sea water	1000–3000	0.001–0.0003
Cooling water (towers)	3000–6000	0.0003–0.00017
Towns' water (soft)	3000–5000	0.0003–0.0002
Towns' water (hard)	1000–2000	0.001–0.0005
Steam condensate	1500–5000	0.00067–0.0002
Steam (oil free)	4000–10,000	0.0025–0.0001
Steam (oil traces)	2000–5000	0.0005–0.0002
Refrigerated brine	3000–5000	0.0003–0.0002
Air and industrial gases	5000–10,000	0.0002–0.0001
Flue gases	2000–5000	0.0005–0.0002
Organic vapors	5000	0.0002
Organic liquids	5000	0.0002
Light hydrocarbons	5000	0.0002
Heavy hydrocarbons	2000	0.0005
Boiling organics	2500	0.0004
Condensing organics	5000	0.0002
Heat transfer fluids	5000	0.0002
Aqueous salt solutions	3000–5000	0.0003–0.0002

Tube Side

- Determine the length and diameter of the tubes according to market standards, referring to Table-3.4.

- Use heuristics to select a standard tube length, typically 6, 8, 12, 16, 20, or 24 feet, ensuring compatibility with industry norms.

Table 3.5 Standard tube diameter values

Outside Diameter (mm)	Wall Thickness (mm)				
16	1.2	1.6	2.0	—	—
20	—	1.6	2.0	2.6	—
25	—	1.6	2.0	2.6	3.2
30	—	1.6	2.0	2.6	3.2
38	—	—	2.0	2.6	3.2
50	—	—	2.0	2.6	3.2

- Select the heat exchanger's construction material.

- Calculate the log mean temperature difference (LMTD) for counter flow provided by,

$$\text{LMTD} = \frac{(T_{h1} - T_{c2}) - (T_{h2} - T_{c1})}{\ln\left(\frac{T_{h1} - T_{c2}}{T_{h2} - T_{c1}}\right)}$$

- Determine the temperature correction factor Ft by using following equations and the figure-1,

$$R = \frac{T_{h1} - T_{h2}}{T_{c2} - T_{c1}}$$

$$S = \frac{T_{c1} - T_{c2}}{T_{h2} - T_{c1}}$$

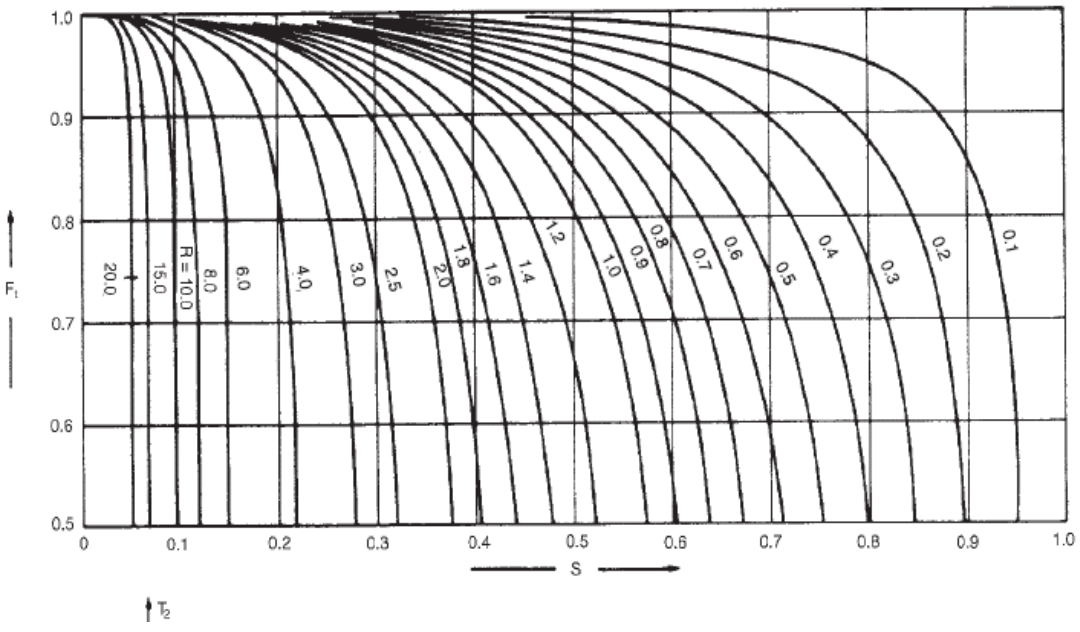


Figure 3.9 Correction factor chart

- Use the equation to calculate overall surface area $Q = UA\Delta T_{LMTD}$,

$$A = \frac{Q}{U\Delta T_m}$$

- To get the number of tubes needed in a heat exchanger, divide the total area (A) by the surface area of each tube.

$$N_{tube} = \frac{A}{\pi dL}$$

Where N_{tube} is the number of tubes, and d and L are the diameter and length of each tube in a single pass.

- Calculate the tube-side velocity (u_t)

$$\text{Tube cross-sectional area } = \pi \times \frac{{d_i}^2}{4}$$

$$\text{Volumetric flow } = \frac{\dot{m}_{in}}{\rho_{in} \times 3600}$$

$$\text{Tube side velocity, } u_t = \frac{\dot{V}}{\text{Area per pass}}$$

- Calculate Reynolds number (Re), Pr and L/d ratio

$$Re_t = \frac{p_i d_i u_t}{\mu}$$

$$Pr = \frac{\mu C_p}{k}$$

$$\frac{L}{d_i} = 406$$

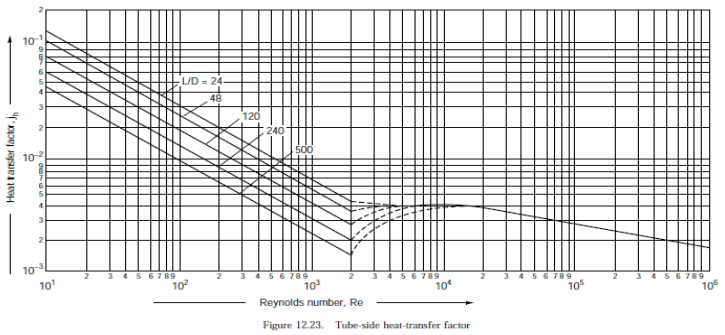


Figure 12.23. Tube-side heat-transfer factor

Figure 3.9 j_h vs Reynold number chart for tube side

Read From fig 2. j_h

- Calculate the Nusselt number

$$Nu = \frac{h_s d_e}{k_f} = j_h Re P^{1/3} \left(\frac{\mu}{\mu_w}\right)^{0.14}$$

- Calculate the tube side film coefficient

$$h_t = \frac{Nu_t k_f}{d_i}$$

- Calculate the pressure drop inside the tube ΔP

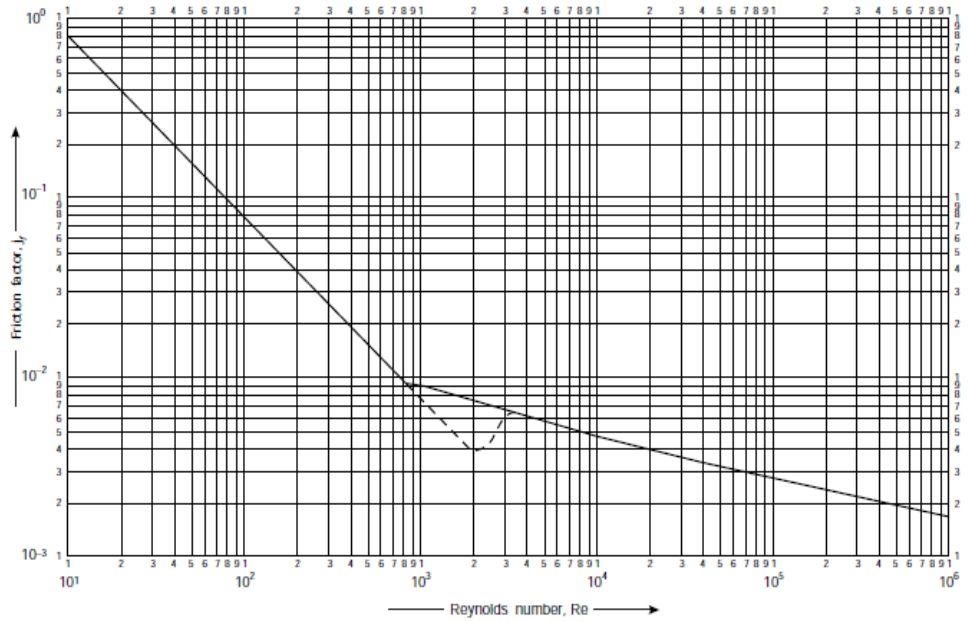


Figure 12.24. Tube-side friction factors
Note: The friction factor j_f is the same as the friction factor for pipes $\phi(= (R/\rho u^2))$, defined in Volume 1 Chapter 3.

Figure 3.10 j_f vs Reynold number chart for tube side

Read j_f from figure 3.10

$$\Delta P_t = N_p * \left[8 j_f \left(\frac{L}{d_i}\right) + 2.5\right] * \frac{\rho v^2}{2}$$

Shell Side

- calculate the pitch of the heat exchanger by,

$$P_t = 1.25 \times do$$

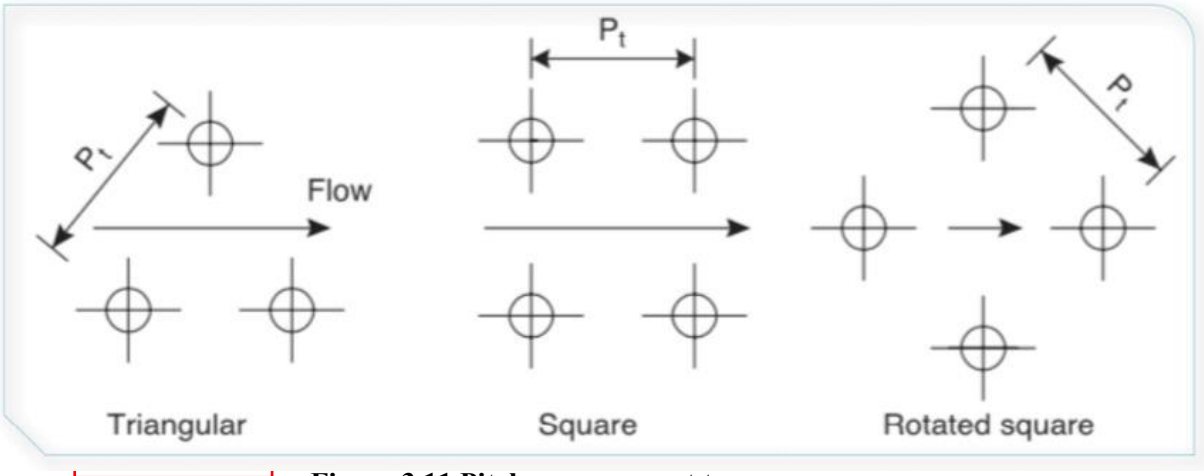


Figure 3.11 Pitch arrangement types

- The bundle diameter can be calculated using (D_b)

$$D_b = d_0 \times \left(\frac{N_T}{K_1}\right)^{\frac{1}{n1}}$$

Constants n_1 and k_1 can be calculated by the table given below,

Table 3.6 values of k and n

Triangular Pitch, $p_t = 1.25d_o$					
No. passes	1	2	4	6	8
K_1	0.319	0.249	0.175	0.0743	0.0365
$>n_1$	2.142	2.207	2.285	2.499	2.675
Square Pitch, $p_t = 1.25d_o$					
No. passes	1	2	4	6	8
K_1	0.215	0.156	0.158	0.0402	0.0331
n_1	2.207	2.291	2.263	2.617	2.643

- Use the figure below to estimate the bundle diameter clearance (BDC).

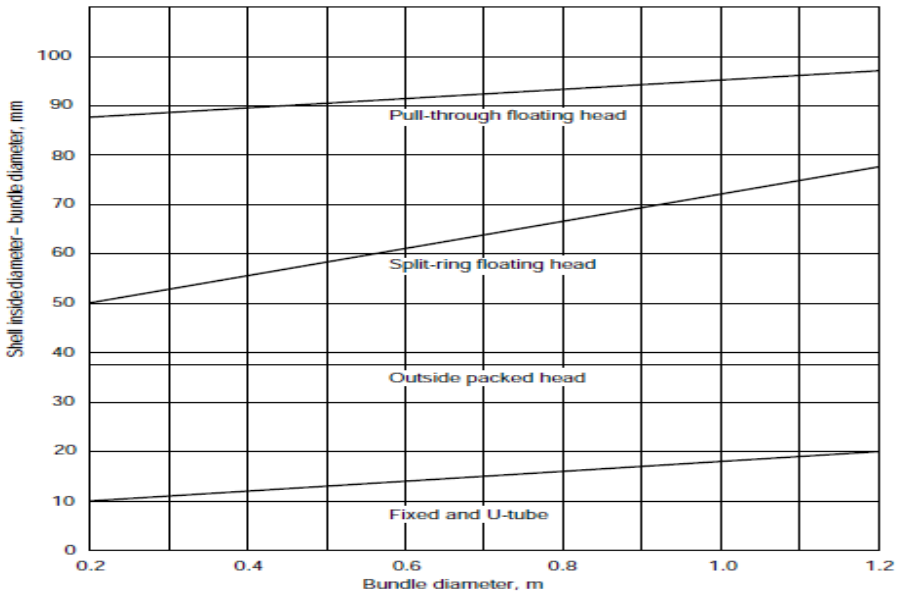


Figure 12.10. Shell-bundle clearance

Figure 3.12 Bundle clearance chart

- Calculate the shell diameter

$$D_s = D_b + BDC$$

- Assume an initial baffle spacing or calculate it using the equation $l_s = {}^{D_s}/_{4}$. Ensure that the pressure drop on the shell side does not exceed 7 psi. If the pressure drop exceeds this value, adjust the baffle spacing until the condition is satisfied.

- Calculate the shell-side crossflow area (A_s)

$$A_s = \frac{(P_t - d_0)}{P_t \times D_s \times l_s}$$

- Calculate the shell-side velocity (u_t)

$$\text{Volumetric flow } = \frac{\dot{m}_{shell}}{\rho_{shell} \times 3600}$$

$$\text{Shell side velocity, } u_s = \frac{\dot{V}}{A_s}$$

- The equivalent diameter of the shell (d_e) for triangular tubes arrangement

$$d_e = \frac{1.1}{d_0}(1.25d_o^2 - 0.917d_0^2)$$

- Calculate the Reynold number, Pr for the shell side and read the graph for 25% baffle cut

$$Re_s = \frac{\rho_s d_e u_s}{\mu}$$

$$Pr_s = \frac{\mu C_p}{k}$$

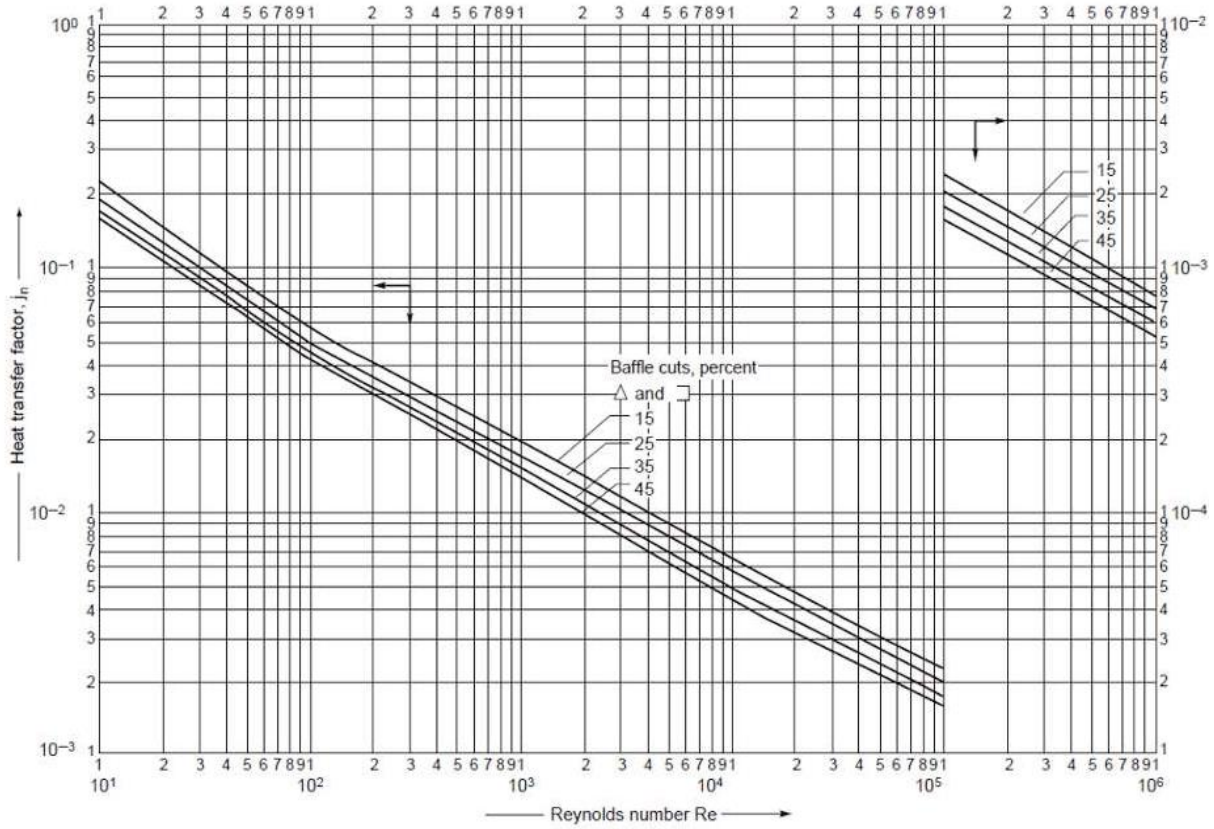


Figure 3.13 j_h vs Reynold number chart for shell side

Read From fig 3.13. j_h

- Calculate the Nusselt number

$$Nu = \frac{h_s d_e}{k_f} = j_h Re P^{1/3} \left(\frac{\mu}{\mu_w}\right)^{0.14}$$

- Calculate the shell side film coefficient

$$h_s = \frac{Nu_s k}{d_e}$$

- Calculate the overall heat transfer coefficient

$$U = \frac{1}{\frac{1}{h_i} + \frac{1}{h_{di}} + \frac{d_i^* \ln\left(\frac{d_0}{d_i}\right)}{2k_w} + \frac{d_i}{d_d h_d} + \frac{d_i}{d_o h_o}}$$

- Calculate the pressure drop inside the shell side ΔP

$$\Delta P_s = \left(8 \times j_f \times \left(\frac{L}{l_B}\right) \times \left(\frac{D_s}{d_e}\right) \times \left(\frac{\mu}{\mu_w}\right)^{-0.14}\right) \times \frac{\rho_s \times v_s^2}{2}$$

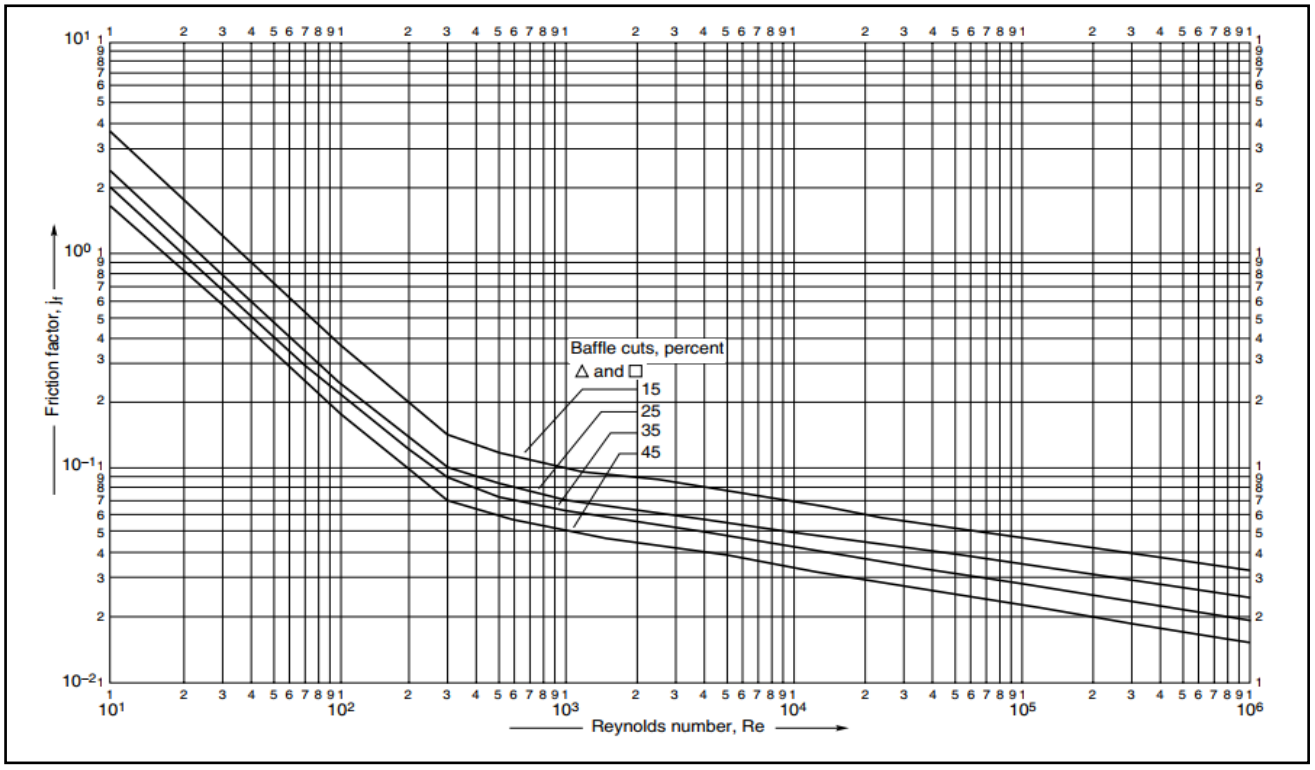


Figure 3.14 j_f vs Reynold number chart for shell side

The term $\left(\frac{\mu}{\mu_w}\right)^{-m}$ may be ignored since it is extremely near to I, and therefore the pressure drop equation occurs.

$$\Delta P_s = \left(8 \times j_f \times \left(\frac{L}{l_B}\right) \times \left(\frac{D_s}{d_e}\right)\right) \times \frac{\rho_s \times v_s^2}{2}$$

Design Calculations

The flow conditions and operational parameters are listed in the table below.

Table 3.7 Properties table

Name	Stream 9	Stream 10	Cooling Water Supply	Cooling Water Return
Temperature (C)	140	40	27	35
Molar flow (kgmole/hr)	348.28	348.28	4023.5	4023.5
Mass flow (Kg/hr)	8373.47	8373.47	72426	72426

Calculation of Heat exchange

To calculate the total heat that need to be exchanged can be calculated by using the sensible heat change formula for the hot gas stream, the average Cp is used.

$$\boldsymbol{Q = \dot{m} C_p \Delta T}$$

$$\boldsymbol{Q = 8373.47 \times 2.9225 \times (140 - 40)}$$

$$\boldsymbol{Q = 679.67\ kW}$$

Assumption of heat transfer coefficient U:

U = 250 W/m².C

Fouling Factor was taken to be 5000 **W/m².C** for the shell side streams while the fouling Factor was taken to be 6000 **W/m².C** for the tube side streams The hot fluid should be placed at the shell side of the heat exchanger and the cold fluid must be placed at the tube side of the heat exchanger.

Tube Side Calculations

Assumptions:

Tube outer diameter = 16 mm,

Wall thickness = 2 mm,

Tube inner diameter = 16 - 2*2 = 12 mm,

tube length = 16ft = 4.8768 m,

Calculation of T_{LMTD}, R, and S

$$LMTD = \frac{(T_{h1} - T_{c2}) - (T_{h2} - T_{c1})}{\ln(\frac{T_{h1} - T_{c2}}{T_{h2} - T_{c1}})} = \frac{(140 - 27) - (40 - 35)}{\ln(\frac{140 - 27}{40 - 35})} = 44.04\ ℃$$

Determine the temperature correction factor Ft by using following equations and the figure-1,

$$R = \frac{T_{h1} - T_{h2}}{T_{c2} - T_{c1}} = 12.5$$

$$S = \frac{T_{c2} - T_{c1}}{T_{h1} - T_{c1}} = 0.0708$$

By using Graph 1:

$F = 0.9$

So the true LMTD will be

$$\Delta T_m = F_t \Delta T_{LMTD}$$

$$\Delta T_m = 0.9 \times 44.04 = 39.64^0 C$$

So, the heat transfer area is

$$A = \frac{Q}{U\Delta T_{\mathrm{m}}} = \frac{679.73\ \mathrm{W} * 1000}{250 \frac{\mathrm{W}}{\mathrm{m^2{}^\circ C}} * 39.64} = 68.6\ \mathrm{m^2}$$

$$N_t = \frac{A}{\pi dL} = \frac{137.2\ \mathrm{m^2}}{\pi * 0.016\ \mathrm{m} * 4.8768\ \mathrm{m}} = 280\ \text{tube}$$

We decided to choose 2 passes.

No of tubes say: 560 tubes

For 2 passes, tubes per pass : 280

Check the tube-side velocity at this stage to see if reasonable

$$m^2 \text{ Tube cross-sectional area } = \pi \times \frac{\mathrm{d_i}^2}{4} = \pi \times \frac{0.012^2}{4} = 0.0001131$$

So, area per pass =280×0.0001131=0.031667 m^2

$$\text{Volumetric flow for water } = \frac{\dot{\mathrm{m}}_{\mathrm{in}}}{\rho_{\mathrm{in}} \times 3600} = 0.020328 \ \frac{m^3}{s}$$

$$\text{Tube side velocity, } \mathrm{u_t} = \frac{\dot{V}}{Area\ per\ pass} = \frac{0.020328}{0.0001131} = 0.64 \frac{m}{s}$$

The velocity is satisfactory since it is between $1 - 2$ m/s

Calculating all the other requirements as listed, we get:

$$Re_t = \frac{p_t d_i u_t}{\mu} = \frac{1000 \frac{\mathrm{kg}}{\mathrm{m}^3} * 0.012 \ \mathrm{m}^* 0.064 \frac{\mathrm{m}}{\mathrm{s}}}{0.0007185 \frac{\mathrm{kg}}{\mathrm{m}^* \mathrm{s}}} = 10721$$

$$Pr = \frac{\mu C_p}{k} = \frac{0.0007185 \times 4.18 \times 10^3}{0.134} = 22.64$$

$$\frac{\mathrm{L}}{d_i} = 406$$

From fig 4.

$$j_h = 0.06$$

$$Nu = \frac{h_s d_e}{k_f} = j_h Re P^{1/3} = 0.06 \times 10721 \times 22.64^{0.33}$$

$$Nu = 1794.95$$

So, the tube side film coefficient is

$$h_i = \frac{Nu_t k_f}{d_i} = \frac{1794.95 \times 0.0003363}{0.012} = 50.30 \ \text{W/m}^2\text{°C}$$

$$\Delta P_t = N_p * \left[8 j_f \left(\frac{L}{d_i}\right) + 2.5\right] * \frac{p u_t^2}{2} = 12 * \left[8 * 0.0174 * \left(\frac{6.096 \text{ m}}{0.0328 \text{ m}}\right) + 2.5\right] * \frac{921.5 \frac{\text{kg}}{\text{m}^3} * \left(0.079 \frac{\text{m}}{\text{s}}\right)^2}{2}$$

$$\Delta P_t = 12392 \text{ Pa} = 1.797 \text{psi}$$

$$\text{Pr}_t = \frac{\mu C_p}{k} = \frac{0.00202 \frac{\text{kg}}{\text{m* s}} * 3590 \left(\frac{\text{j}}{\text{kg}} \text{°C}\right)}{0.132 \text{ W/m°C}} = 54.93$$

Shell-side

Doing the same calculations as per the procedure, we get bundle diameter:

$$\text{Pt} = 1.25 * \text{do} = 1.25 * 16 \text{ mm} = 20 \text{ mm}$$

$$D_b = d_0 * \left(\frac{N_t}{K_1}\right)^{\frac{1}{n_i}} = 16 \text{ mm} * \left(\frac{280}{0.249}\right)^{\frac{1}{2.207}} = 386 \text{ mm}$$

Using Fig-3, Bundle Diameter Clearance $= 20$ mm

Diameter of the shell:

$$D_s = D_b + BDC = 386 + 20 = 406 \text{ mm}$$

is assumed to be $D_s/4l_s$

$$A_s = \frac{(P_t - d_0) * D_s l_s}{Np * P_t} = \frac{(20 - 16)\text{mm} * 406 \text{ m} * 101 \text{ m}}{0.1 * 20 \text{ mm}} = 0.08236 \text{ m}^2$$

$$G_s = \frac{m_s}{\rho} = \frac{8373.4 \frac{\text{kg}}{\text{h}}}{1.734 \frac{\text{kg}}{\text{m}^3}} = 1.341 \frac{\text{m}^3}{\text{s}}$$

$$u_s = \frac{G_s}{A_s} = \frac{1.341 \frac{\text{m}^3}{\text{s}}}{0.08236 \text{ m}^2} = 16.282 \frac{\text{m}}{\text{s}}$$

$$d_e = \frac{1.1}{d_0}(P_t^2 - 0.785 d_0^2) = \frac{1.1}{16 \text{ mm}} * ((20 \text{ mm})^2 - 0.785(16 \text{ mm})^2) = 0.01136 \text{ m}$$

$$\text{Re} = \frac{\rho u_s d_e}{\mu} = \frac{1.734 \times 16.282 \times 0.01136}{1.55 \times 10^{-5}} = 20735.36$$

$$\mathrm{Pr}_s = \frac{\mu C_p}{k} = \frac{2.92 \times 10^3 \times 1.55 \times 10^{-5}}{0.132} = 0.34$$

$$Nu_s = j_h \,\mathrm{Re}_s \, Pr_s^{\frac{1}{3}} = 0.1^{*}20735.36 * 0.34^{0.33} = 1456$$

$$h_o = \frac{Nu_s k_f}{d_e} = \frac{1456^{*}0.05103\ \mathrm{W/m^{\circ}C}}{0.01136\ \mathrm{m}} = 6540\ \mathrm{W/m^2{}^{\circ}C}$$

$$U_i = \frac{1}{\frac{1}{h_i} + \frac{1}{h_{di}} + \frac{d_i^{*}\ln\left(\frac{d_0}{d_i}\right)}{2k_w} + \frac{d_i}{d_0 h_{do}} + \frac{d_i}{d_0 h_0}}$$

$$U_i = \frac{1}{\frac{1}{50.30} + \frac{1}{5000} + \frac{0.012^{*}\ln\left(\frac{0.016}{0.012}\right)}{2 * 33} + \frac{0.012}{0.016^{*}6000} + \frac{0.012}{0.016 * 6540}} = 147\,\frac{\mathrm{W}}{\mathrm{m^2C}}$$

$$\%\ \text{error} = \frac{|250 - 147|}{\left(\frac{250 + 146}{2}\right)} * 100\% = 51\%$$

Pressure Drop in shell side

The pressure drop inside the shell side of the heat exchanger is given by

Using Fig 5,

$$\mathrm{J_f} = 0.9$$

$$\Delta P_s = 8 j_f \left(\frac{D_s}{D_e}\right)\left(\frac{L}{l_b}\right) * \frac{\rho u_s}{2} * \frac{1}{Np}$$

$$= 8 * 0.9 * \left(\frac{0.406\ \mathrm{m}}{0.001136\ \mathrm{m}}\right) * \left(\frac{4.8768\ \mathrm{m}}{0.101\ \mathrm{m}}\right) * \frac{1.734\,\frac{\mathrm{kg}}{\mathrm{m^3}} * 16.28\,\frac{\mathrm{m}}{\mathrm{s}}}{4}$$

$$\Delta P_s = 87290.76\ Pa = 0.873\ bar$$

Pressure Drop in tube side

The pressure drop inside the tube side of the heat exchanger is given by

Using Fig5 ,

$$\mathrm{J_f} = 0.02$$

$$\Delta P_t = N_p * \left[8 j_f \left(\frac{L}{d_i}\right) + 2.5\right] * \frac{\rho u_t^2}{2}$$

$$= 12 * \left[8 * 0.02 * \left(\frac{4.8768\ \text{m}}{0.012\ \text{m}}\right) + 2.5\right] * \frac{1000\ \frac{\text{kg}}{\text{m}^3} * \left(0.64\ \frac{\text{m}}{\text{s}}\right)^2}{2}$$

$$\Delta P_t = 27824.05\ \text{Pa} = 0.278\ \text{bar}$$

Table 3.8 H-104 Specification Sheet

Equipment Name	Cooler		
Objective	To cool down the product gases produced from Butanol Reactor (R-101)		
Equipment Number	H-104		
Type/Reason	Shell and Tube, because flow rate is high. So, Area is greater than $3m^2$		
Location	After Ethanol pre-heater (H-103) and before flash separator (F-101)		
Material of Construction	Stainless Steel		
Insulation	Mineral Wool		
Shell Side			
Inlet temperature (°C)	140	Outlet temperature (°C)	40
Tube Side			
Inlet temperature (°C)	27	Outlet temperature (°C)	35
Tube diameter (mm)	16	Tube length (m)	4.8768
Number of Tube Passes	2	Number of Tubes	560
Tube bundle Diameter (mm)	386	Shell Diameter (mm)	406
Q_{total} (KW)	679	LMTD (°C)	39.63
U ($W/m^2/°C$)	146	Heat Exchanger Area (m^2)	68.6

3.2.3 Two phase separator

Introduction

In a butanol production plant, phase separation and distillation are essential steps to ensure selective recovery of n-butanol and efficient recycling of unreacted components. The separator (F-101) plays a key role in separating hydrogen from the reaction mixture, while the downstream distillation section recovers ethanol and purifies butanol.

Physical properties

The physical properties of feed and product is shown in table 9.

Table 3.9 Physical properties of feed and product

Stream Number	Density (kg/m^3)	Viscosity (cP)	Heat Capacity (kJ/kgmol.C)
10	1.681	1.87	84.24
15	831.3	3.071	146.5
16	7.839e-002	9.12e-003	28.46

Chemical design calculations

Designing a flash column in the chemical industry involves several steps and considerations. Let's go through some calculations for designing a flash column for hydrogen separation using distillation as the separation technique.

Phase Separator Design:

For secondary separation, maximum allowable vapor velocity is calculated by using the Souders-Brown equation.

$$v = K * \sqrt{\frac{\rho L - \rho V}{\rho V}}$$

$$v = K * \sqrt{\frac{831.3 - 0.07839}{0.07839}}$$

Where, ρ_L, ρ_V are liquid and vapor densities. K is calculated as follows.

$$K = 0.35 - \frac{0.01(P - 100)}{100} = 0.35 - \frac{0.01(14.7 - 100)}{100} = 0.35706\ ft/s$$

Where P is in psig, K is in feet/sec.

For no mist separator multiply by 0.01

K=0.0035706 ft./s = 0.0010888 m/s

$$v = 0.0010888 * \sqrt{\frac{831.3 - 0.07839}{0.07839}} = 0.112\ \frac{m}{s}$$

Vessel diameter is calculated based on vapor velocity V.

$$D = \sqrt{\frac{4Q}{v\pi}}$$

Where Q (vapor flow rate) =174 kmol/hr = 5.2 m^3/hr or 0.00144 m^3/s

$$D = \sqrt{\frac{4 * 0.00144}{0.112 * \pi}} = 0.127m$$

$$\text{Liquid volumetric flow-rate } = \frac{8018}{3600 \times 831.3} = 2 \times 10^{-6}\ \text{m}^3/\text{s}$$

Allow a minimum of 10 minutes' hold-up.

$$\text{Volume held in vessel } = 2 \times 10^{-6} \times (10 \times 60) = 1.6 \times 10^{-3}\ \text{m}^3$$

$$\text{Liquid depth required, } h_v = \frac{\text{Volume held} - \text{up}}{\text{Vessel cross sectional area}}$$

$$h_v = \frac{1.6 \times 10^{-3}}{(\pi \times 0.127^2/4)} = 0.126\ m$$

Increase to 0.13 m to allow space for positioning the level controller.

Total separator height (L) $= 0.4 + D_v + \frac{D_v}{2} + h_v$

Total vessel height (L) $= 0.4 + 0.127 + \frac{0.127}{2} + 0.13 \ = 0.720$ m

$$V = \frac{\pi}{4} d^2 h = \frac{\pi}{4} * 0.127^2 * 0.72 = 1.92\ m^3$$

The vessel thickness is calculated using the formula below:

$$\mathbf{t = \frac{P \times r_i}{SE_j - 0.6P} + C_c}$$

$P = 1.2 \times$ Operating pressure $= 1.2 \times 101.325 = 121.59$ kPa (20% was added for safety purposes)

$$\mathbf{r_i = \frac{D_V}{2} = \frac{3.31}{2} = 1.653\ m}$$

S = 72400 kPa

Taken from the below table for the stainless-steel 304.

$\mathbf{E_j}$ = Fractional joint efficiency

$\mathbf{C_c}$ = 0.003175 m

$$\mathbf{t = \frac{P \times r_i}{SE_j - 0.6P} + C_c = \frac{121.59 \times 1.653}{72400 \times 0.88 - 0.6 \times 121.59} + 0.003175}$$

$$\mathbf{= 0.006\ m = 6\ mm}$$

The calculated wall thickness is 6 mm.

Results of chemical design calculation

The volume of flash column comes out to be 1.92 m^3.The flash column is an important part of the butanol production process, ensuring effective separation of light components such as hydrogen and unreacted ethanol from the heavier alcohols. The separation is achieved by reducing the operating pressure, allowing the lighter components to vaporize while butanol and higher alcohols remain in the liquid phase. The flash column volume is determined using the Souders–Brown equation, which for this system comes out to be 1.92 m^3. To minimize operational risks, suitable materials of construction must be selected to resist solvent attack, and corrosion prevention measures such as protective linings or cathodic protection are required.

Table 3.10 Specification Sheet for F-101

Equipment Name		Two Phase Separator	
Objective		To remove hydrogen from the Alcohols and water stream	
Type		Vertical Column Separator	
Location		After H-104	
Material of Construction		Stainless Steel	
Insulation		None	
Operating Condition			
Operating Temperature (°C)	40	Operating Pressure (bar)	1.013
Design Considerations			
Liquid Density (kg/m^3)	831	Gas Density (kg/m^3)	0.078
Liquid hold up time (min)	10	Liquid depth (m)	0.943
Gas Flow rate (kg/hr)	710	Liquid Flow rate (kg/hr)	16036
Demister (Yes/No)	NO		
Dimensions			
Diameter (m)	1.096	Height (m)	3

3.2.4. Distillation Column (D-104)

Introduction

The main objective of column D-104 is to separate Butanol from Hexanol and Octanol, where the Butanol (top stream) is the final product with 97% purity. The separation tower is operated under the assumption of 97 % Butanol in the top stream.

Process variables:

- Feed, distillate, and residue flow rate ($\frac{kmol}{hr}$)
- Relative volatility
- Reflux ratio

Design criteria:

- 97% Butanol in the top

Assumptions:

- multicomponent system
- The column operates at 1.2 bar = 1.2 atm.
- Efficiency of the column 70%

Material of construction*: Stainless Steel*

Table 3.2 Material balance on the column results.

Component	Feed (F)	Distillate (D)	Residue (B)
Flow rate ($\frac{Kmol}{hr}$)			
Butanol	77.007	77.007	0
Hexanol	1.914	1.439	0.474
Octanol	1.914	0.226	1.688
Total	80.83	78.675	2.162
Component Mole Fraction (X_F, X_D, X_B)			
Butanol	0.9526	0.9788	0.0000
Hexanol	0.0237	0.0183	0.2193
Octanol	0.0237	0.0029	0.7807
Total	1	1	1

Fractional recovery helps identify the light and heavy key components in a distillation process. The method involves several steps:

$$\text{FR} = \frac{\text{Flow out}}{\text{Flow in}}$$

FR of Butanol at the distillate:

(NK) $$\text{FR} = \frac{77.007\ \text{kmol/hr}}{77.007\text{kmol/hr}} = 1$$

FR of Hexanol at the Top:

(LK) $$\text{FR} = \frac{1.439\ \text{kmol/hr}}{1.914\ \text{kmol/hr}} = 0.7521$$

FR of Octanol at the Bottom:

(HK) $$\text{FR} = \frac{1.688\ \text{kmol/hr}}{1.914\ \text{kmol/hr}} = 0.8819$$

Where Octanol is HK since its boiling point is higher than Hexanol while the Butanol is non-key component because it does not distribute itself in two stream instead all butanol goes up.

Material Balance

With the assumption of constant molal overflow, the liquid and vapor amounts in both the rectifying and stripping sections are determined through a material balance, and the results are summarized in Table 3.11.

Rectifying section:

$V_0 = L_0 + D$

- V_0 is the total amount of vapor at top section $\left(\frac{Kmol}{hr}\right)$.
- L_0 is the amount of liquid retained to the column $\left(\frac{Kmol}{hr}\right)$.
- D is the distillate molar flow $\left(\frac{Kmol}{hr}\right)$.

Reflux ratio $= \frac{L_0}{D} =$

$= \times\ D = \frac{Kmol}{hr} L_0 = \times\ D$

$= \frac{Kmol}{hr} V_0 = +$

Stripping section:

$$L_n = F + L_0$$

$$V_n = L_n - B$$

- L_n is the total amount of liquid at bottom section of the column $\left(\frac{Kmol}{hr}\right)$.
- F is the amount of feed $\left(\frac{Kmol}{hr}\right)$.
- V_n is the amount of vapor at the bottom section $\left(\frac{Kmol}{hr}\right)$.
- B is the bottom product $\left(\frac{Kmol}{hr}\right)$.

$$L_n = \ + = \frac{Kmol}{hr}$$

$$V_n = - = \frac{Kmol}{hr}$$

Bubble Point Temperature Calculations

- Feed Temperature

$$ln(p^{sat}) = A - \frac{B}{T + C}$$

The Antoine Equation is employed to determine the vapor pressure (p^{sat}) of each component. Were,

- p^{sat} is the saturated vapor pressure in (atm).
- T is the temperature in (K).
- Constants A, B, and C, which are specific to each component, are listed in Table 4.3.2.

Table 3.3 Antoine equation constants

	A	B	C
Hexanol	4.014	1362.1	230.12
Octanol	3.74844	1196.64	149.043

Raoult's law is then applied to verify the guessed temperature by comparing the partial pressures of each component in the liquid phase to the corresponding vapor pressures obtained through the Antoine Equation.

$$y_i\, P = \, x_i\, p^{sat}$$

For the bubble point calculations $\sum y_i = 1$

at the guessed temperature, T=118 ℃ + 273.15= 391.15 KP^{sat}

$$Ln(P^{sat}_{\text{HEXANOL}}) = 4.014 - \frac{1362.1}{391.15 + 230.12} = 1.821$$

$$P^{sat}_{\text{HEXANOL}} = 6.17\ atm$$

$$Ln(P^{sat}_{\text{OCTANOL}}) = 3.7844 - \frac{1196.64}{391.15 + 149.043} = 1.57$$

$$P^{sat}_{\text{OCTANOL}} = 4.87\ atm$$

Mole fraction:

$$y_{\text{HEXANOL}} = \frac{x_{\text{HEXANOL}} * P^{sat}_{\text{HEXANOL}}}{P} = \frac{(0.0237)(6.17\ atm)}{1.46\ atm} = 0.1217$$

$$y_{\text{OCTANOL}} = \frac{x_{\text{OCTANOL}} * P^{sat}_{\text{OCTANOL}}}{P} = \frac{(0.0237)(4.87\ atm)}{1.46\ atm} = 0.0948$$

$$\sum y_i = 0.1217 + 0.0948 = 0.2165$$

Since $\sum y_i \neq 1$ at the guessed temperature, iterations are required in order to find the bubble point temperature of the feed. Aspen plus is used and the bubble point found to be 410 K. The feed section analysis resulted in indicating a saturated liquid feed with q=1, since $T_{Bubble} = T_{Feed}$.

Residue temperature

The design pressure of the residue section, 2.2 atm, is used to calculate the temperature at which the first vapor bubble forms upon heating, essentially determining the bubble point of the residue stream. Similar methods and calculations were applied as in the feed section analysis, where the bubble point temperature was determined to be 558 K.

Distillate temperature

The design pressure of the distillate section is assumed to be atmospheric, at 110 kPa, and is used to determine the temperature at which saturated vapor begins to condense, known as the dew point. For the dew point calculation, a similar procedure was followed as in the feed section analysis; however, the liquid mole fraction was used in Raoult's law. Using Aspen plus again, the distillate temperature was found to be 391 K, with an operating pressure of 110 kPa.

Relative Volatility

Antoine equation was used to calculate the pressure at the saturation temperature of the top and bottom of the column. The distillate section's saturated pressure was found at 391 K, while the bottom section's saturated pressure was found at 558 K. The average relative volatility was found to be higher than 1, which indicates that a distillation column can be used for this separation.

$$\alpha = \frac{P^{sat}_{HVC}}{P^{sat}_{LVC}} = \frac{P^{sat}_{OCTANOL}}{P^{sat}_{HEXANOL}}$$

$$\alpha_{TOP} = \frac{P^{sat}_{HEXANOL}}{P^{sat}_{OCTANOL}} = \frac{0.2496atm}{0.0629\ atm} = 3.968$$

$$\alpha_{Bottom} = \frac{P^{sat}_{HEXANOL}}{P^{sat}_{OCTANOL}} = \frac{24.58\ atm}{11.39\ atm} = 2.065$$

$$\alpha_{avergae} = \sqrt{\alpha_{Bottom} * \alpha_{TOP}} = \sqrt{(3.968)(2.065)} = 2.862$$

Vapor liquid equilibrium data

The equilibrium data (VLE data) for the distillation system was obtained using Aspen plus at a pressure of 1.46 atm, as presented in Table 3.13 Here, y_{eq} and x_{eq} represent the vapor and liquid mole fractions of Hexanol at equilibrium, respectively. This data is also displayed graphically in Figure.

Table 3.4 VLE data for Hexanol & Octanol system.

Xeq	Yeq
0	0
0.02	0.0499638
0.08	0.185959
0.1	0.227072
0.18	0.373287
0.2	0.405748
0.22	0.436737
0.3	0.547419
0.38	0.639822
0.4	0.660466
0.48	0.734547
0.5	0.751134
0.58	0.810758
0.6	0.824128
0.68	0.872238
0.7	0.883036
0.78	0.9219
0.8	0.930623
0.9	0.969045
0.98	0.994337
1	1

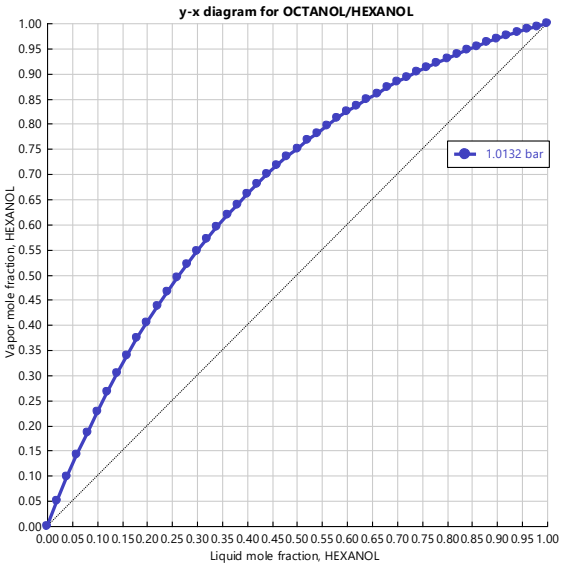


Figure 3.1 VLE data for H_2O- MA mixture system

Number of Equilibrium stages (Number of theoretical plates)

The Fenske equation is given by

$$N = \frac{\log\left[\left(\frac{X_d}{1-X_d}\right)\left(\frac{1-X_b}{X_b}\right)\right]}{\log\ \alpha_{avg}}$$

Here

$X_d = 0.018$

$X_b = 0.219$

$\alpha_{avg=2.862}$

$$N = \frac{\log\left[\left(\frac{1}{1-0.018}\right)\left(\frac{1-0.219}{0.219}\right)\right]}{\log\ 2.862} = 5.74$$

Or more precisely 6 number of stages.

Minimum Reflux ratio (R_{min})

q-line at the feed properties

q indicates the quality of the feed which is the heat required to vaporize one mole of feed over the molar latent heat of the feed, also represented by the equation below.

$$q = \frac{H_v - H_F}{H_v - H_L}$$

- H_v is the molar enthalpy of saturated vapor
- H_L is the molar enthalpy of saturated liquid
- H_F is the molar enthalpy of the feed at entering conditions.

- For a saturated liquid feed: $H_F = H_L$

$$q = \frac{H_v - H_F}{H_v - H_L} = 1$$

Now apply the formula below we can find θ

$$\frac{\alpha_A x_{fA}}{\alpha_A - \theta} + \frac{\alpha_B x_{fB}}{\alpha_B - \theta} = 1 - q$$

$$\frac{3.968 \times 0.0237}{3.968 - \theta} + \frac{2.065 \times 0.0237}{2.065 - \theta} = 1 - 1$$

Apply the solver function we get

$\theta = 1.4$

Now apply the formula for R_m

$$\frac{\alpha_A x_{dA}}{\alpha_A - \theta} + \frac{\alpha_B x_{dB}}{\alpha_B - \theta} = R_m + 1$$

$$\frac{3.968 \times 0.018}{3.968 - 1.4} + \frac{2.065 \times 0.219}{2.065 - 1.4} = R_m + 1$$

$$R_m = 1.041$$

Optimum reflux ratio (R)

Optimum or economical reflux ratio is 1.2 − 1.5 times the minimum reflux ratio. In this design, 1.5 times the minimum reflux ratio is used.

$$R_{opt} = 1.5 \times R_m = 1.5 \times 1.041 = 1.561$$

$$R_{opt} = 1.561$$

Actual number of stages:

The overall efficiency can be calculated based on the following relation:

$$\mathrm{E_0} = 0.52782 - 0.2751 \log(\mathrm{k\mu}) + 0.044923(\log(\mathrm{k\mu}))^2$$

Where

μ = Viscosity of liquid in cP

k = volatility

With the viscosity $\mu = 0.93$ cP and the relative volatility $(k) = 4.11$.

$$\mathrm{E_0} = 0.52782 - 0.16 + 0.01521 = 0.38303$$

The overall efficiency would be used. Now, we can relate the actual number of trays to the theoretical number of stages by the following:

$$\text{Overall efficiency} = \frac{\text{Theoretical number of stages}}{\text{Actual number of stages}}$$

$$\text{Actual number of stages} = \frac{\text{Theoretical number of stages}}{\text{Overall efficiency}}$$

$$\text{Actual number of stages} = \frac{6}{0.383} = 16 \text{ stages required}$$

Column height

According to design heretics, most towers use tray spacing of 0.6 m (24 in). As well as clearances of 1.2 m and 1.5 m both at the top and bottom of the column, respectively.

This allowance is provided for the liquid at the bottom and the vapor disengaging space at the top.

$$\mathrm{H} = (\text{ number of trays } - 1) \times \text{ tray spacing } + \text{ allowances}$$

So, putting values

$$\mathrm{H} = (\ 16\ - 1) \times\ 0.6 + (1.2{+}1.5)$$

$$\mathrm{H} = 11.7\ \mathrm{m}$$

Density of Vapor stream

The vapor is assumed to be ideal gas, and under this assumption the ideal gas law is used.

$$\rho_v = \frac{PM}{RT}$$

ρ_v represents the density of the vapor in $\frac{g}{m^3}$ P is the pressure in atm, M is the molar mass in $\frac{g}{\mathrm{mol}}$, R is the universal gas constant, $8.205745 \times 10^{-5}\ \frac{\mathrm{m^3\ atm}}{\mathrm{mol.K}}$. The tower operates under pressure of 1.1 atm, the temperature of the vapor stream at the top of the tower is 391 K .

$$M = (MW_{Hexanol})(x_{Hexanol}) + (MW_{\mathrm{Butanol}})(x_{\mathrm{Butanol}}) + (MW_{\mathrm{Octanol}})(x_{\mathrm{Octanol}})$$

$$M = (0.02499)\left(\frac{\mathrm{g}}{102\ \mathrm{mol}}\right) + (0.97)\left(74\,\frac{\mathrm{g}}{\mathrm{mol}}\right) + (0.0049)\left(130\,\frac{\mathrm{g}}{\mathrm{mol}}\right) = 75.1\,\frac{\mathrm{g}}{\mathrm{mol}}$$

After putting values we get

$$\rho_v = \frac{PM}{RT} = \frac{(1.1\ \mathrm{atm})\left(75.1\,\frac{\mathrm{g}}{\mathrm{mol}}\right)}{\left(8.205745 \times 10^{-5}\,\frac{\mathrm{m^3 \cdot\ atm}}{\mathrm{mol} \cdot\ \mathrm{K}}\right)(391\ \mathrm{K})} = 2576.6\,\frac{\mathrm{g}}{\mathrm{m^3}} * \frac{1\ \mathrm{kg}}{1000\ \mathrm{g}}$$

$$\rho_v = 2.576\,\frac{\mathrm{kg}}{\mathrm{m^3}}$$

Gas Velocity

$,u = \frac{F_s}{\sqrt{\rho_v}}$

Where

$$F_s = 1.2 - 1.5\frac{m^3}{s}$$

So calculate the gas velocity for both values of F_s

$$u = \frac{F_s}{\sqrt{\rho_v}} = \frac{1.2}{\sqrt{2.576\frac{\text{kg}}{\text{m}^3}}} = 0.7475\frac{\text{m}}{\text{s}}$$

And

$$u = \frac{F_s}{\sqrt{\rho_v}} = \frac{1.5}{\sqrt{2.576\frac{\text{kg}}{\text{m}^3}}} = 0.9344\frac{\text{m}}{\text{s}}$$

So the average velocity can be calculated by

$$\text{Average Velocity} = \frac{u_1+u_2}{2} = \frac{0.7475\frac{\text{m}}{\text{s}}+0.9344\frac{\text{m}}{\text{s}}}{2} = 0.841\frac{\text{m}}{\text{s}}$$

So the column diameter is calculated.

Column Diameter

$$D(u_{\text{avg}}) = \sqrt{\frac{4\dot{V}}{u \times \pi}}$$

Where

is the volumetric flow rate of the vapor phase in the distillation column $\dot{V} = \frac{\dot{m}}{\rho_v}\dot{V}$

Reflux Ratio $= \frac{L_0}{D} = 1.561$

$$L_0 = 1.561 \times D = 1.561 \times 78.675 \frac{\text{kmol}}{\text{hr}} = 122.81 \frac{\text{kmol}}{\text{hr}}$$

So

$$\text{V}_0 = L_0 + \text{D} = 122.81 \frac{\text{kmol}}{\text{hr}} + 78.675 \frac{\text{kmol}}{\text{hr}} = 201.485 \frac{\text{kmol}}{\text{hr}}$$

The volumetric flow rate will be

$$\dot{V} = \frac{\dot{m}}{\rho_v} = \frac{\left(201.485 \frac{\text{kmol}}{\text{hr}}\right) * \left(75.1 \frac{\text{kg}}{\text{kmol}}\right)}{2.576 \frac{\text{kg}}{\text{m}^3}} = 5874.038 \frac{\text{m}^3}{\text{hr}} * \frac{1\text{hr}}{3600\text{sec}} = 1.613 \frac{\text{m}^3}{\text{sec}}$$

The column diameter is

$$D(u_{\text{avg}}) = \sqrt{\frac{4\dot{V}}{u \times \pi}} = \sqrt{\frac{(4)\left(1.613 \frac{\text{m}^3}{\text{sec}}\right)}{\left(0.841 \frac{\text{m}}{\text{s}}\right) \times \pi}} = 1.571 \text{ m}$$

Column Area

$$A_c = \left(\frac{\pi D^2}{4}\right) = \left(\frac{(\pi)(1.571m)^2}{4}\right) = 1.233\ m^2$$

Cross sectional area of the downcomer $(A_d) = 18\ \% \times A_c$

$A_d = 0.18 \times 1.233$

$A_d = 0.221\ m^2$

Net area available for vapor-liquid release $(A_n) = Ac - Ad$

Net area available for vapor-liquid release $(A_n) = 1.233 - 0.211 = 1.011\ m^2$

Active area of the plate $(A_a) = A_c - 2A_d$

$A_a = 1.233 - 2 \times 0.221$

$A_a = 0.811\ m^2$

Area of the holes $(A_h) = 5\%$ o f A_a

$A_h = 0.05 \times 0.811 = 0.04\ m^2$

Number of holes

Diameter of hole $= d_h = 15\ mm$

$$\text{Area of one hole} = \frac{\pi \times {d_h}^2}{4} = \frac{\pi \times (0.015)^2}{4} = 1.766 \times 10^{-4}\ m^2$$

$$\text{Number of holes per one plate} = \frac{A_h}{\text{Area of one hole}}$$

$$\text{Number of holes per one plate} = \frac{0.04}{766 \times 10^{-4}} = 229 \text{ holes}$$

Tower Volume

$$V = \left(\frac{\pi D^2}{4}\right) H = \left(\frac{(\pi)(1.571m)^2}{4}\right)(11.7) = 22.66\ m^3$$

With the design margin 20% $V = 28\ m^3$

Material Construction

Material of construction used here will be stainless steel 304 due to its resistance against corrosion. Design Pressure: The equipment working pressure highest is 2.2 bar. Keeping a margin of 20% our design pressure is 2.64 bar.

Design Temperature: The equipment working temperature highest is 285ºC. Keeping a margin of 20% our design temperature is 342ºC.

Material of Construction

The best material for the absorber is stainless steel because feed might be fouling and corrosive. It is therefore more costly than carbon steel but resistant to corrosion.

$$t = \frac{P \times r_i}{SE_j - 0.6P} + C_c$$

= efficiency of joints expressed as fraction (take it 0.88)E_j

= allowance for corrosion (m) (take 0.00318 inch)C_c

Operating pressure = 146 kPa

P = 1.2*146 = 175 kPa

$$r_i = \frac{D_c}{2} = \frac{2.22}{2} = 1.11 \text{ m}$$

S = maximum allowable working stress = 72400 kPa (For stainless steel 304)

= Efficiency of joints expressed as fraction = 0.88E_j

= allowance for corrosion = 0.00318 mC_c

$$t = \frac{175 \times 1.11}{72400 \times 0.88 - 0.6 \times 175} + 0.00318 = 0.006234 \text{ m} = 6.234 \text{ mm}$$

The calculated value of the thickness is smaller than the practical value of the vessel thickness. It can be compared from the table given below:

Table 3.5 Thickness Data

Vessel diameter (m)	Minimum thickness (mm)
1	5
1 to 2	7
2 to 2.5	9
2.5 to 3.0	10
3.0 to 3.5	12

Hence, the minimum thickness will be taken from the table and it is given below:

Minimum practical thickness of the vessel (for diameter less than 2.22 m)

= 9 mm

Table 3.6 Specification Sheet of D-104

Equipment Name	Distillation Column		
Objective	The objective of this distillation column is to separate the Butanol from heavier alcohols like Hexanol and Octanol.		
Equipment Number in Aspen Plus	D-104		
Type/Reason	Trayed type column, because the calculated diameter of the column is greater than 0.6 m, which makes it trayed column.		
Location	It is located after the Distillation column D-101, and it is the last column of the sheet.		
Material of Construction	Stainless steel (304) was used.		
Dimensions			
Column Diameter (m)	1.571	Column Height (m)	11.7
Number of Trays	16	Reflux Ratio	1.561
Tray Spacing (m)	0.6	Type of tray	Sieve trays
Diameter of hole (mm)	15	Number of holes/trays	702

3.3. Equipment Sizing

3.3.1 Sizing of Pump P-101

The function of pump is simply to move the fluid by increasing the pressure. In our process the pump P-101 is responsible for pumping ethanol to Butanol production process.

According to 1st Law of thermodynamics

$$\frac{dU}{dt} = \Sigma F_i \hat{H}_i + Q + W_s - P\frac{dV}{dt}$$

For steady-state, adiabatic, and single stream:

$\frac{dU}{dt} = \frac{dV}{dt} = 0$ For steady state system the accumulation term is zero

$Q = 0$ For adiabatic system

$$W_s = H_f - H_i$$

For isentropic and reversible process, we further have:

$$W_s(\text{ isentropic }) = \Delta H = \int_{P_i}^{P_f} VdP$$

As the liquid stream volume, V is independent of pressure. Thus,

$$W_s(\text{ isentropic }) = V(P_f - P_i)$$

Hence, the shaft power of the pump with efficiency is:

$$W_s = \frac{W_s(\text{ isentropic })}{\eta}$$

The values from the material balance is

$V = 10.18\ \frac{m^3}{hr}$

$P_f = 5000$ kPa

$P_i = 101$ kPa

Now put the vales in the formula we get the final answer

$$W_s(\text{ isentropic }) = \frac{10.18\ m^3}{hr} * (5000 - 101) * 10^3 * \frac{1}{3600} * \frac{1}{1000} = 13.852\ kW$$

By assuming that $\eta = 0.75$, we get the shaft power to be:

$$W_s = \frac{13.852}{0.75} = 18.471\ kW$$

Pump Selection

It is an important to choose the right type of pump to deliver the work required. In P-100, the moving fluid is the Ethanol which is considered low-viscous fluid. Since the mentioned fluid is of low viscosity and relatively high flowrate, the suitable pump to be used is the centrifugal pump. Centrifugal pumps are characterized by their specific speed.

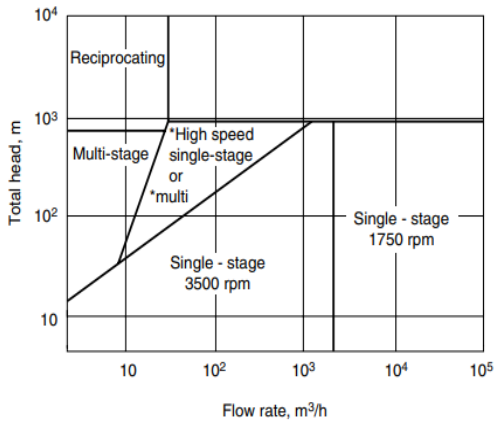


Figure 3.2 Selection of Centrifugal Pump

So the pump P-101 is single-stage centrifugal pump.

3.3.2 Sizing of Pump (P-104)

The function of pump is simply to move the fluid by increasing the pressure. In our process the pump P-104 is responsible for pumping ethanol and water mixture to column D-102.

According to 1st Law of thermodynamics

$$\frac{dU}{dt} = \Sigma F_i \hat{H}_i + Q + W_s - P\frac{dV}{dt}$$

For steady-state, adiabatic, and single stream:

$\frac{dU}{dt} = \frac{dV}{dt} = 0$ For steady state system the accumulation term is zero

$Q = 0$ For adiabatic system

$$W_s = H_f - H_i$$

For isentropic and reversible process, we further have:

$$W_s(\text{ isentropic }) = \Delta H = \int_{P_i}^{P_f} VdP$$

As the liquid stream volume, V is independent of pressure. Thus,

$$W_s(\text{ isentropic }) = V(P_f - P_i)$$

Hence, the shaft power of the pump with efficiency is:

$$W_s = \frac{W_s(\text{ isentropic })}{\eta}$$

The values from the material balance is

$V = 1.97\ \frac{m^3}{hr}$

$P_f = 300$ kPa

$P_i = 101$ kPa

Now put the vales in the formula we get the final answer

$$W_s(\text{ isentropic }) = \frac{1.97\ m^3}{hr} * (300 - 101) * 10^3 * \frac{1}{3600} * \frac{1}{1000} = 0.021\ kW$$

By assuming that $\eta = 0.75$, we get the shaft power to be:

$$W_s = \frac{0.021}{0.75} = 0.028\ kW$$

Pump Selection

It is an important to choose the right type of pump to deliver the work required. In P-104, the moving fluid is the Ethanol and water mixture which is considered low-viscous fluid. Since the mentioned fluid is of low viscosity and relatively high flowrate, the suitable pump to be used is the centrifugal pump. Centrifugal pumps are characterized by their specific speed.

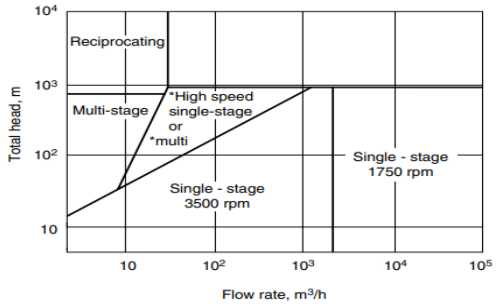


Figure 3.17 Selection of Centrifugal Pump

So, the pump P-104 is single-stage centrifugal pump.

Chapter 4

Process Control and instrumentation

4.1. Types of Controllers

Flow Controller

The speed of the feed in the processing unit is controlled by these controllers. By far, the most popular type of flow rate sensor is the cleft panel. Most of the time, the vent panels are made to let water pressure drop by 20 to 200 inches. It is also possible to use adventure lines and turbine counters.

Temperature Controller

Most of the time, thermocouples are used to measure temperature. It has two separate lines that make a voltage difference when they are exposed to different temperatures.

Pressure Controller

To measure and find difference pressure, you can use Bordon tubes, bellows, and a diaphragm. So, in a mechanical system, the movement of the spring keeps the crushing force of the process stable. The process pressure can be linked to the position of the spring.

Level Controller

The Liquid level can be detected in a various way. The three among them are most common,

- proceeding the location of a float, that is lighter than the fluid.

- Assess the superficial weight of a heavy object as it buoyed up more or less by the liquid. (these are called displacement meters).

- Computing the divergence in static pressure between two fixed advancements, one in the vapor side above the liquid and the other one which is under the liquid surface. The differential pressure between two-level wires is directly related to the liquid level in the container.

4.2.Various types of supervision Loops

There are three major types of supervision courses of action:

- Feedforward control manipulations

- Feedback control manipulations

- Inferential control manipulations

Feedforward supervision manipulation

The configuration of supervision in which the disturbance enters the system, propagates throughout the system, and is detected in the system's output. The controller corrects the effect by making adjustments to the system after it has been detected, which is referred to as a feedback control scheme.

Feedback supervision manipulation

Feedforward control schemes are characterized by the initial detection of the control configuration in a disturbance by the measuring instrument, followed by the correction of the system's conditions by the controller to ensure that the combined effect of the disturbance and system remains at the set point prior to entering the system.

Inferential supervision manipulation

Control schemes that modify parameter values through the utilization of second-class measurements. Main objective of this scheme is to maintain the controlled status of unmeasured variables.

4.3. Types of Feedback Controller

The control modes of various feedback controllers are used to estimate the intended type of counter. In other words, these actions demonstrate the console's functionality, which is the coordination of the output signal with the input signal or inaccuracy. The three fundamental control modes are as follows:

- Proportionate controller (P controller)

- Proportionate Integral controller (PI controller)

- Proportionate Integral derivative controller (PID controller)

on industrial scale righteously integral, proportional, or derivative controllers uncommonly occur alone in the supervision system.

4.4. Instrumentation

Instruments are the measuring devices that are to be used to obtain the most precise value of the process variable. The fundamental requirement for the regulation of a process is the measurement of its parameters. The current process industry is reliant on the ability to take repeatable and accurate measures in order to manage the process more efficiently and effectively. This measurement action is executed using measuring instruments. In some cases, the measurement is obtained through indirect methods, in which one parameter is measured in relation to other parameters, which are then calibrated with the required parameter. For instance, our control objectives may occasionally fall under the category of unmeasurable outputs. The reason for this may be the unavailability of the measuring instrument. For instance, the use of composition measuring devices is prohibitively expensive, as they typically rely on spectrometric devices for measurement, a cost that only a small number of industries can afford. In such instances, we estimate other variables that are readily calculable. Secondary measurements are the term used to describe these types of measurements. In the same vein, we can assume that the temperature parameter of any source must be measured using the most well-known and basic equipment, the mercury glass thermometer. When the bulb of the thermometer is exposed to the source in question, the fluid expands as a result of heat transfer from the source to the bulb glass, and subsequently from the glass wall to the mercury fluid. The volume of the fluid

increases as it expands, and the volume is ultimately calibrated in terms of temperature, which can be measured. In the same vein, numerous examples may be cited. The primary method of measurement involves the conversion of one form of energy into another. The four parameters that are most frequently measured in any process industry are pressure, level, temperature, and flow. These four parameters accounted for 80% of the measurements. Other parameters, such as density and pace, comprise the remaining 20%.

Types of Instrumentation

There are three types of instrumentation:

- Recording instrumentation.

- Indication instrumentation.

- Alarm instrumentation

Recording Instrumentation

The type in which temperature, pressure, and flow are continuously measured and all values of these parameters are recorded over time. Followed by the continuous display of these measurements in a chart that pertains to the time of instrumentation recording. These charts can be administered by a time clock operation and may be circular or linear in nature.

Indication Instrumentation

Measurements conducted in the field and shown via dials or similar devices are referred to as indicated measurements from indication instruments. This instrumentation is beneficial for field operators. Although it is a continuous measurement, the values are recorded and shown only at the moment of measurement. Currently, digital displays are employed for these indications.

Alarm Instrumentation

Instruments utilized to identify hazardous or worrisome situations are referred to as alarm instrumentation. These devices stay inactive under standard operating settings; nevertheless, when parameters reach specified hazardous limits, they activate alarms and indicate detection. Switches are primarily utilized for this form of detection, activating under frightening conditions and remaining inactive during normal conditions.

Control Scheme of Reactor

Objective

To control the reactor outlet temperature.

Manipulated Variables

Control the flow rate of the Hitec fluid at the shell side of the reactor.

Loads or Disturbances

Following are typical disturbances

- Flow rate of feed

- Composition of feed

- Level of the water at the shell side of the salt cooler

- Temperature of feed

- Inlet temperature of the Hitec fluid

Control Looping

The system consists of four control loops which control

- The reactor feed rate

- The reactor outlet temperature

- The coolant supply temperature

- The salt cooler level

Reactor Feed Rate Controller

The reactor feed rate will be controlled using a simple PI controller. To minimize disturbances in the reactor and downstream units, set-point pacing will be applied. An air-to-open control valve will be used so that, in the event of a failure in the plant's instrument air system, the valve will automatically close. This safety feature will help prevent a runaway reaction and return the plant to a safe condition during an emergency.

Reactor Outlet Controller

The reactor outlet temperature will be controlled using combined feedback–feedforward strategy. Feedforward control will detect and respond to variations in the reactor feed rate before they affect the temperature control loop, while feedback control will maintain the temperature set point accurately and eliminate steady-state offset, compensating for any inaccuracies in the feedforward model. A cascade control loop will also be incorporated to improve the system's dynamic response, ensuring stable and efficient operation of both the reactor and the overall plant. The temperature of the reaction mixture within the reactor cylinders will be monitored through an auctioneering control loop, which will receive signals from at least 15 temperature sensors positioned throughout the reactor. If the maximum reactor temperature exceeds the process limit, the water supply to both the salt-cooler and coolant control valves will automatically open fully. Both of these valves will be air-to-close (fail-open) type to ensure reliable cooling in case of an air supply failure.

Coolant Inlet Temperature Controller

The water flow to the salt chiller will be used to regulate the coolant supply temperature entering the reactor. To achieve more accurate control of the salt inlet temperature, the salt-cooler pressure controller can be integrated into a cascade control system. However, this configuration may transmit disturbances to the downstream power generation equipment, potentially causing adverse effects on other process units.

Water Level Controller

A feedback control strategy will also be applied in the fourth control loop. This controller will adjust the steam generation rate in response to changes in the water supply and maintain the desired water level in the salt-cooler. To further enhance the inherent safety of the system, an air-to-close control valve will once again be utilized.

Positioning of the Indicators

Pressure indicators will be installed at the gas inlet and outlet to monitor the pressure differential across the catalyst and detect any abnormal increases. A high-pressure alarm will require immediate operator action and may trigger an emergency shutdown if necessary. A pressure indicator will also be installed on the salt cooler to provide early warning of potential safety valve activation and steam release.

Temperature sensors will be placed at various points along the reactor tubes to measure temperature distribution and identify the maximum temperature, as tubes near the reactor core may behave differently from those near the outer wall.

Coolant temperatures will also be monitored to prevent excessive salt temperatures that could hinder temperature control and create undesirable reaction conditions. Low- and high-temperature alarms will alert operators to overcooling or overheating, helping prevent solidification or salt–metal interactions at high temperatures.

The complete control configuration of the reactor is illustrated in the figure below:

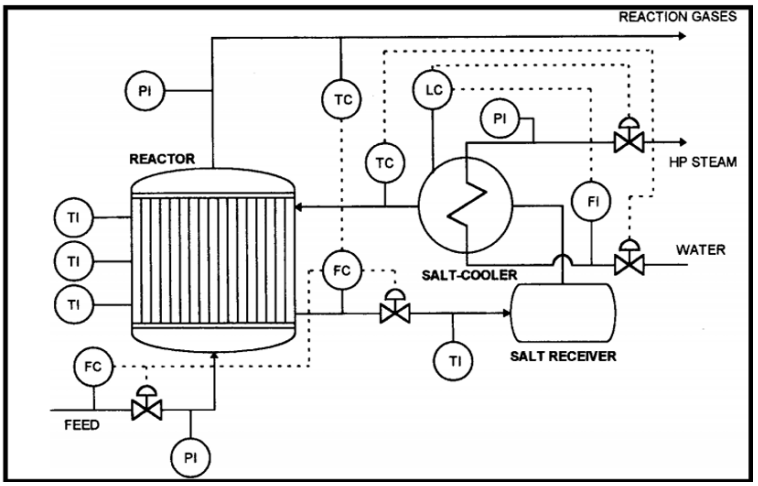


Figure 4.1 Control Scheme of Reactor

Chapter 5

Hazard and Environmental Analysis

5.1. HAZOP Analysis

Purpose

HAZOP (Hazard and Operability) study is the technique that is applied to determine the hazards and operability problems with process streams for ethanol to butanol using the Guerbet reaction. The nodes of emphasis are the storage of feeds, pretreatment of feeds, reactor(s), product split (distillation/flash), heat integration (reboilers, condensers), and dispatch/storage.

Parameters

Following enlisted parameters are considered while conducting HAZOP Study:

- Study Nodes**:** The location at which parameters are investigated for deviation.
- Intention: It defines how the plant is expected to operate in the absence of deviation.
- Deviations: The way process departs from design expectation.
- Causes: The reason why deviation might occur.
- Consequences: The ultimate results of the deviations.
- Safeguards**:** Facilities that help to reduce the deviation.

Guided Words

Table 5.1 guided words used in HAZOP analysis

	Guide Words	Details
1	No	Negation of design intend
2	Less	Quantitative Decrease
3	More	Quantitative Increase
4	Part of	Qualitative Decrease
5	As well as	Qualitative Increase
6	Reverse	Logical opposite of the intend
7	Other Than	Complete Substitution

HAZOP Process

HAZOP is performed using a standardized procedure:

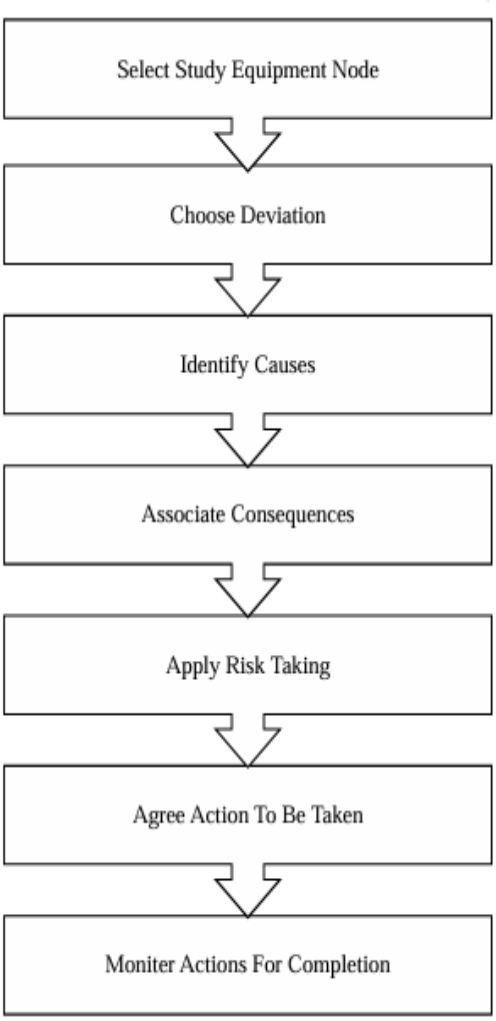


Figure 0 HAZOP Process Steps

Representative HAZOP

Node/Equipment	Parameter	Deviation	Possible Causes	Consequences	Existing Safeguards	Recommended Actions
Feed Tank & Transfer Pump	Flow	No Flow	Pump failure, valve closed, line blockage	Reactor starved, temperature rise in heater, process interruption	Level indicator, flow switch, low-flow alarm	Provide standby pump, auto switch-over, interlock to stop feed heater
	Flow	More Flow	Pump speed malfunction, control valve failure	Reactor overpressure, high conversion, catalyst overheating	Flow control valve, PID controller	Install flow limiter, high-flow alarm
	Composition	Wrong Feed Composition	Contamination, wrong ethanol batch	Off-spec product, catalyst poisoning	Feed quality check, inline analyzer	Include prefeed sampling, raw material specification check
Reactor (Guerbet Reaction Unit)	Temperature	High Temperature	Cooling failure, runaway reaction, faulty thermocouple	Catalyst deactivation, side reactions, fire risk	Temperature controller, high-temp trip, pressure relief valve	Provide redundant temperature sensors, emergency quench system
	Pressure	High Pressure	Outlet valve closed, gas generation, fouling	Reactor rupture, safety valve lifting	Pressure control valve, PSV, rupture disc	Add secondary PSV, ensure regular vent inspection
	Mixing	Poor Agitation	Agitator motor fault, low RPM	Hot spots, incomplete reaction, yield loss	Vibration monitor, current sensor	Include RPM interlock, maintenance schedule

Heat Exchanger / Condenser	Temperature	Low Outlet Temp	Fouling, cooling water overflow	Incomplete vaporization or condensation	Temperature indicator, flow control	Schedule cleaning, maintain heat transfer coefficient
	Pressure	Tube Leak	Corrosion, mechanical failure	Cross contamination between hot/cold fluids	Pressure alarm, leak detection	Periodic NDT inspection, corrosion allowance material
Distillation Column (Butanol Recovery)	Reflux Ratio	Low Reflux	Pump failure, valve malfunction	Low purity butanol, higher VOC emissions	Level & flow indicators, composition control	Install reflux pump standby, online purity analyzer
	Temperature	High Bottom Temperature	Reboiler duty too high	Overheating, product degradation	Temperature controller, reboiler interlock	Add alarm & auto shutdown on over-temp
	Level	High Bottom Level	Poor draw-off, control valve failure	Flooding, carryover to overhead	Level controller, alarm	Install high-high level interlock
Product Storage Tank	Pressure	Overpressure	Inert gas overfill, thermal expansion, vent blocked	Tank rupture, VOC release	Pressure relief vent, flame arrestor	Regular vent inspection, install pressure transmitter
	Leakage	-	Flange failure, corrosion, gasket wear	Fire, toxic release, environmental contamination	Bund wall, gas detector	Double-seal gaskets, preventive maintenance
Wastewater Collection Tank	COD / Organic Load	High Concentration	Process upset, leakage of product	Overload of treatment plant, environmental	COD monitoring, pH control	Segregate high-strength streams, equalization tank

5.2. Environmental Considerations

This section covers likely environmental impacts and how to control them in a Guerbet-based butanol plant.

Airborne Emissions

Potential Pollutants

- VOCs: unreacted ethanol, produced butanol, higher alcohols, light organics from side reactions.

-Combustion emissions: CO_2, NO_x, CO, particulate from boilers or flares.

-Fugitive emissions: pump seals, valves, tank vents.

-Oduor: volatile organic Oduors from storage and wastewater.

Controls & Management

-Use closed systems for raw material handling and product transfer; minimize open venting.

-Install condensers and solvent recovery for vents; use carbon adsorption or thermal VOC oxidizer for low-flow high-VOC streams.

-Implement flare only as emergency relief; minimize routine flaring.

-Leak detection & repair (LDAR) program for rotating equipment and joints.

-Stack monitoring for combustion sources; continuous emissions monitoring where required by permits.

Monitoring

- Periodic VOC stack sampling, continuous hydrogen/flammable gas detectors in plant zones, and CO_2/NO_x monitoring on boilers.

Waterborne Emissions (Wastewater)

Sources

- Process water, column bottoms with organics, condensate from vapour recovery, cooling tower blowdown, storm water contaminated by spills.

Pollutant Types

- High COD/BOD (organic alcohols), dissolved organics (alcohols, aldehydes), salts/neutralization chemicals, spent solvents, traces of catalyst components.[4]

Treatment Strategy

1. **Segregation & Equalization**: Separate high-strength streams (e.g., from product recovery) and neutral/low-strength streams. Use an equalization tank with sampling port.
2. **Physical Separation**: Oil/water separators, decanters to remove free-phase organics.
3. **Biological Treatment**: Sequencing batch reactor (SBR) or activated sludge for biodegradable organics (ethanol/butanol are biodegradable). Ensure acclimation and tunable retention time based on COD.
4. **Polishing**: Activated carbon adsorption for refractory organics, air stripping for volatile alcohols if necessary, and disinfection.
5. **Sludge Handling**: Dewater sludge from biological units and arrange proper disposal or use as fuel if permitted.

Monitoring & Discharge

- Monitor COD, BOD_5, total suspended solids (TSS), pH, temperature, and specific organics. Discharge to municipal works or surface water must comply with permit; otherwise, consider zero-liquid discharge (ZLD) or reuse options.

Solid Waste

Sources

- Spent catalyst, filter cakes, spent activated carbon, contaminated rags/absorbents, process residues.

Handling & Disposal

- Classify waste as hazardous or non-hazardous based on local regulations and leachable metals or organic concentration.

- Spent catalysts typically require regeneration or specialized hazardous waste disposal; analyze metals and organics before disposal.

- Implement waste minimization (catalyst life optimization, solvent recovery) and a documented waste handling plan with licensed haulers.[5]

5.3. Inherent Safety Aspects & Emissions Table

Inherent Safety Principles Applied

- **Minimize**: Reduce inventory of flammable feed stocks at any single location by using smaller tanks and frequent feed.

- **Substitute**: Where possible, use less hazardous solvents or process aids.

- **Moderation**: Operate reactors at the lowest feasible temperature/pressure consistent with conversion.

- **Simplify**: Avoid unnecessary piping and complex valve networks that increase leak points.

Emissions & Allowable Limits

Table 5.3 emission and allowable limit for the butanol production

Pollutant	Source	Typical Concentration	Local Limit (KSA)	Recommended Control	Monitoring Frequency
VOCs (total)	Reactor vents, distillate vents, storage	20–50 mg/Nm³	≤ **50 mg/Nm³**	Condenser → solvent recovery → carbon bed/thermal oxidizer	Continuous / weekly checks
Butanol (as VOC)	Distillation overheads	—	≤ **50 mg/Nm³**	Overhead condensers, return to process or recovery	Online GC or periodic sampling
Ethanol (VOCs)	Purge streams	—	≤ **50 mg/Nm³**	Closed-loop recovery, adsorption	Periodic
CO_2, NO_x, CO	Boilers, heaters	—	NO_x ≤ **150 mg/Nm³**	Efficient combustion, low-NO_x burners, SCR if needed	Continuous emissions monitoring (if required)
COD / BOD	Wastewater	50–250 mg/L	BOD_5 ≤ **100 mg/L**, COD ≤ **250 mg/L**	Equalization → biological treatment → polishing	Daily / weekly
TSS	Wastewater	—	TSS ≤ **100 mg/L**	Clarifiers, filters	Daily / weekly

Regulatory Compliance & Permitting

- The authoritative body in the Kingdom is the Ministry of Environment, Water and Agriculture (MEWA) through the Executive Regulations on Air Quality.
- Permit applications must have emission inventories, evaluations of worst-case scenarios, mitigation strategies, and continuous monitoring plans.
- The Royal Commission for Jubail and Yanbu, along with MEWA, establishes criteria for the treatment and disposal of wastewater, where relevant.
- Emergency response planning, stakeholder participation over odor and noise, and documentation of emissions and fugitive leaks are obligatory.

Chapter 6

Economic Analysis

6.1 Capital cost Estimation

Butanol has proven to become a vital alternative fuel and chemical feedstock in order to ensure the sustainable development of the industrial sectors. This has been attributed to the fact that butanol has high energy density, compatibility with the existing fuel infrastructure, as well as lower volatility when compared to ethanol. In the present petrochemical industry process, butanol can essentially be obtained from the hydrocarbons obtained from naphtha. This procedure enables the effective use of the existing refinery streams to develop important oxygenates that would greatly benefit the energy as well as chemical sectors.

The capital costs related to the main process equipment such as reactors, distillation columns, heat exchangers, and process pumps as well as utilities form the main basics involved in determining the economic viability of butanol production. To realistically estimate the equipment costs associated with the butanol production process, it is imperative to obtain credible information regarding the capital costs of equipment used in the butanol production process. This characteristic is very important because equipment cost-related information has the ability to accurately represent the total fixed capital investment cost of the plant. In addition, the equipment cost-related information has the ability to give a clear estimation of the production costs and, once again, total profit from the process of producing butanol.

Equipment Cost

Using the Bare Module Cost (BMC) method, the total investment includes not only the main equipment (like reactors, distillation columns, and pumps) but also related installation and process costs. This method provides a realistic estimate by accounting for materials, design, fabrication, and transportation. Cost estimates follow industry standards and are adjusted using the Chemical Engineering Plant Cost Index (CEPCI) of 803 to reflect current economic conditions.

Equipment Name	Equipment Type	Purchased Equipment Cost ($)	Bare Module Cost ($)
C-101	Centrifugal	$ 606,000	$ 3,480,000
H-101	Fixed, Sheet, or U-Tube	$ 53,600	$ 330,000
H-102	Fixed, Sheet, or U-Tube	$ 50,900	$ 314,000
H-103	Fixed, Sheet, or U-Tube	$ 47,700	$ 294,000
H-104	Floating Head	$ 53,400	$ 329,000
H-105	Floating Head	$ 62,200	$ 288,000
M-101	Impeller	$ 44,300	$ 66,500
M-102	Impeller	$ 44,300	$ 66,500
P-101	Centrifugal	$ 14,400	$ 71,400
P-102	Centrifugal	$ 5,700	$ 28,300
P-103	Centrifugal	$ 11,800	$ 58,500
P-104	Centrifugal	$ 4,960	$ 24,600
R-101	PFR	$ 29,100	$ 43,700
D-101	Sieve Trays Column	$ 149,000	$ 694,000
D-102	Sieve Trays Column	$ 37,400	$ 141,000
D-103	Sieve Trays Column	$ 45,400	$ 200,000
D-104	Sieve Trays Column	$ 37,400	$ 197,000
F-101	Vertical	$ 9,660	$ 76,400

Table 6.1 Equipment Cost

The Total Bare Module Cost (BMC) is crucial for estimating capital investment, assessing economic feasibility, and calculating ROI in a butanol production plant. Since equipment costs form a large part of fixed investment, accurate BMC projections support better financial planning, design optimization, and cost management, ensuring realistic budgeting and long-term profitability.

Fixed Capital Cost

The fixed capital cost includes all expenses for purchasing, installing, and commissioning the main and supporting equipment in the Guerbet reaction–based butanol plant. Equipment costs were estimated using bare module factors, industry data, and cost correlations adjusted for design and operating conditions. The total Fixed Capital Investment (FCI) was obtained by summing the Bare Module Costs (BMC) of all key units, excluding land, working capital, and startup costs. CAPCOST software and CEPCI data were used for accurate and updated cost estimation. Additional costs such as auxiliary facilities, contractor fees, and contingencies were included using the Total Module Factor (FM) to ensure realistic project evaluation.

$$C_{TM} = 1.18 \sum C_{BM} = \$\,7910000$$

The butanol plant's capital cost includes installation, equipment, and all supporting facilities needed for safe and efficient operation. This covers infrastructure, buildings, materials, and utility systems like steam, electricity, and water. Including these ensures a complete financial plan for the plant. The total facility cost combines the base module costs with additional expenses for processing units and related equipment.$C_{GR} = 0.5 \sum C_{BM}^{\circ} + C_{TM} = \$\,9650000$

Working Capital Cost

The working capital cost for the butanol plant includes expenses for raw materials, labor, utilities, and waste management. Raw material costs cover ethanol, catalysts, and chemicals used in the Guerbet reaction. Labor costs include operator wages, supervision, and support staff. Utilities such as electricity, steam, and water ensure continuous and reliable operations. Waste management expenses cover the treatment of gases, effluents, and catalyst disposal to meet environmental standards. These components are vital for sustainable operation and profitability. Butanol, the main product, sells for about $1.40–$1.60 per kilogram depending on purity and market conditions..

Table 6.2 working capital cost

Material Name	Price ($/kg)	Classification	Flowrate (kg/h)	Annual Cost ($)
Ethanol	$ 2.70	Reactant	2977.20	$ 56745221
Butanol	$ 1.5	Product	5710	$ 71277930

Name	Utility Used	Actual Usage	Annual Utility Cost (AED)
H-104	Cooling Water	1350 MJ/h	$ 4000
H-105	Cooling Water	22020 MJ/h	$ 6000
R-101	Custom	500 MJ/h	$ 4990
P-101	Electricity	52.9 kilowatts	$ 26,600
P-102	Electricity	4.29 kilowatts	$ 2,160
P-103	Electricity	35.7 kilowatts	$ 18,000
P-104	Electricity	0.529 kilowatts	$ 266

Table 3 delineates the annual utility expenses associated with the hydrogen manufacturing process, with the overall cost approximated at $76,216. Table 4 illustrates the anticipated revenue from sales. The total cost, comprising operating labor, utility, and raw material expenses, is determined using Equation 3:

Revenue From Sales	$	71,277,930
C_{RM} (Raw Materials Costs)	$	56,745,221
C_{UT} (Cost of Utilities)	$	76,200
C_{WT} (Waste Treatment Costs)	$	452,384
C_{OL} (Cost of Operating Labor)	$	105,800

As shown in **خطأ! لم يتم العثور على مصدر المرجع.**, the projected revenue from sales. This cost, which is the sum of operating labor cost, utilities costs, raw material costs etc. is calculated using Equation 3

Total Investment

An investment in butanol production plant fixed capital cost (FCC) includes both the fixed capital cost and the working capital (WC) for optimal operation of the facility. The fixed capital cost includes process equipment, including both existing and new plant infrastructures. Fixed capital that supports the function includes utility and auxiliary systems that are considered necessary to operate the facility.
Working capital (WC) accounts for all raw materials, utilities, labor pay, maintenance supplies, and other short-term operational necessities until the plant becomes

stabilized and flow of revenue, and is typically estimated as a fraction of the fixed capital cost based on a method and based on the industry. Therefore, total investment (TI) for the plant can be written as:

$$\text{Total Investment} = \text{Fixed Capital cost} + \text{Working capital cost}$$

$$\text{Total Investment} = \$\ 16300000$$

6.2 Economic Analysis

Selling Price Estimation

The economic viability for the butanol production plant is heavily influenced by the selling price of the products produced. The market prices for Butanol, Hexanol, and Octanol were analyzed from the commodity indexes and recent industrial pricing. The reflected selling prices for Butanol, Hexanol, and Octanol are $1.45/kg, $2.10/kg, and $2.80/kg, respectively. These prices reflect the present state of the petrochemical market from which they were derived and presumed production-grade purity. The pricing was based on anticipated product yield, capacity, and expected conversion efficiency to determine anticipated revenue for the plant as a whole. Setting profit margins and the breakeven point depends on careful pricing estimates that make sure production costs can be compared to those of competitors in the market and that cash flow forecasts are accurate.

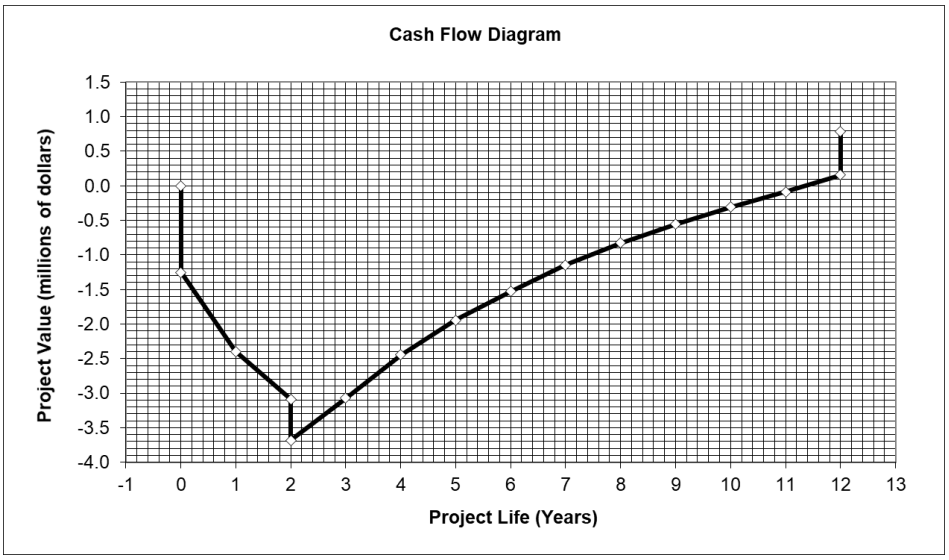
Estimation of Economic Indicators

To see if the project is financially feasible, metrics like Net Present Value, Internal Rate of Return, and Payback Period are looked at. A discounted cash flow analysis is done to account for the time value of money, and a sensitivity analysis is done to see how changes in key parameters would affect profits.

Discounted cash flow diagram

The cash flow diagram illustrates the project's financial viability using the discounted cash flow (DCF) method, which considers the time value of money. During the first two years, project value drops due to capital and working capital investments made during construction. After this phase, positive cash flows begin, and the project value rises. The breakeven point, where cumulative discounted cash flow becomes positive, occurs at about 5.3 years. Afterward, profits continue to grow, confirming the project's financial feasibility. The final year includes the salvage value, which adds to total long-term profitability.

Figure 6.1 Discounted cash flow diagram



Non-discounted cash flow diagram

The cash flow diagram shows the project's non-discounted cash flow over its lifetime. In the first two years, cash flow is negative due to capital and construction expenses. After construction, revenue begins to exceed costs, leading to positive cash flow. The project reaches breakeven in about 4.5 years and then continues to generate increasing profits. In the final year, the salvage value adds to the total project value. This analysis evaluates profitability without adjusting for time or discounting factors.

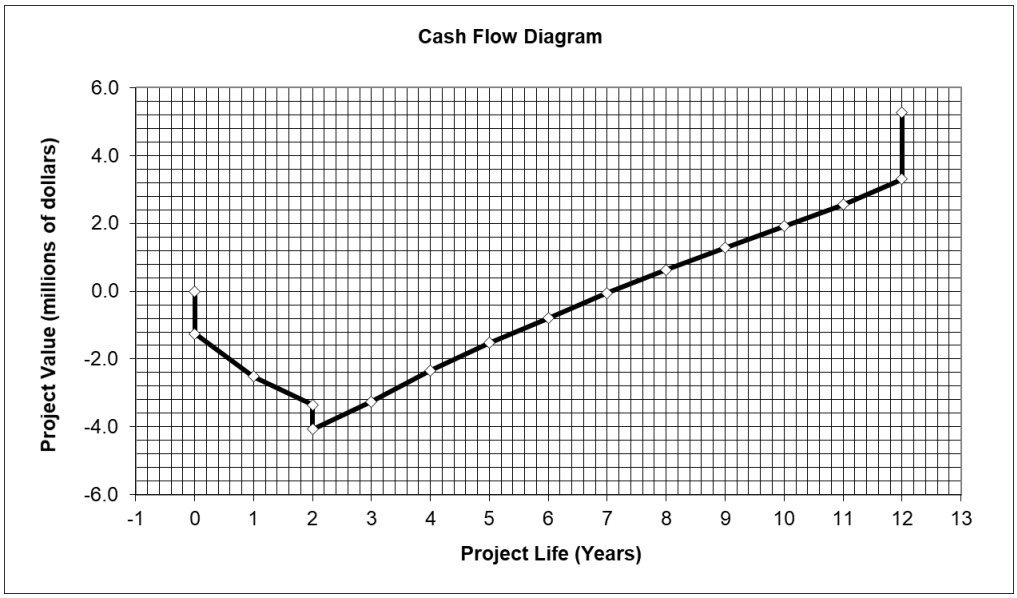


Figure 6.3 Non-discounted cash flow diagram

The main difference between discounted and non-discounted profitability criteria lies in the time value of money. Discount methods reduce the value of future cash flows, while non-discounted methods treat them at full value. As a result, the discounted NPV (0.78 million) is lower than the non-discounted cumulative cash (5.27 million). Similarly, the discounted rate of return (13.54%) is less than the non-discounted rate (25.12%). The payback period is also longer for the discounted method (3.2 years) compared to the non-discounted one (2.5 years) because discounting lowers the value of future cash flows.

Discounted Profitibility Criterion	
Net Present Value (millions)	0.78
Discounted Cash Flow Rate of Return	13.54%
Discounted Payback Period (years)	3.2

Non-Discounted Profitibility Criteria	
Cumulative Cash Position (millions)	5.27
Rate of Return on Investment	25.12%
Payback Period (years)	2.5

Figure 6.4 Discounted and non-discounted profitability Criterion

Since the NPV is positive, and the rate of return is reasonable, the project appears **feasible.**

Net present value

Figure 7 shows the cumulative distribution of Net Present Value (NPV), a key financial measure that accounts for the time value of money. A higher NPV indicates a more profitable project. The graph ranges from a low NPV of about -6.5 million dollars (worst-case loss) to a high NPV of about 8.5 million dollars (maximum potential gain).

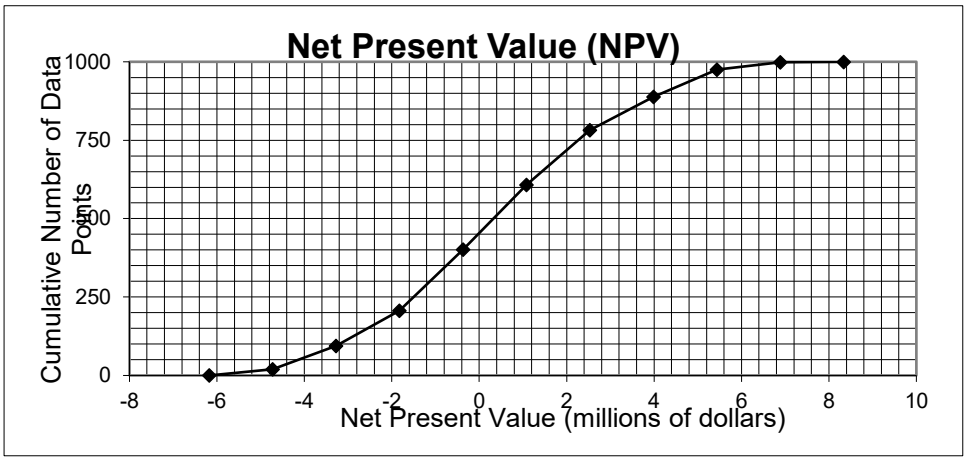


Figure 6.5 NPV curve data

About half of the scenarios show a positive NPV, meaning the project has nearly equal chances of profit or loss. The curve's turning point is between $0 and $2 million, and it flattens beyond $6 million, indicating few highly profitable cases. Negative NPV areas represent financial losses, while positive areas suggest that the project can be economically feasible under favorable conditions.

Discounted Cash Flow Rate of Return Data

The picture below is associated with the Discounted Cash Flow Rate of Return (DCFROR) graphic. The DCFROR represents the interest rate at which the net present value (NPV) is exactly zero**.** The lowest value for the DCFROR inthe data set is 0.01, which means the least amount of return. The highest value for the DCFROR in the data set is 0.40, which means the highest return. The range of expected rates of return on investment is set by the difference between the readings.

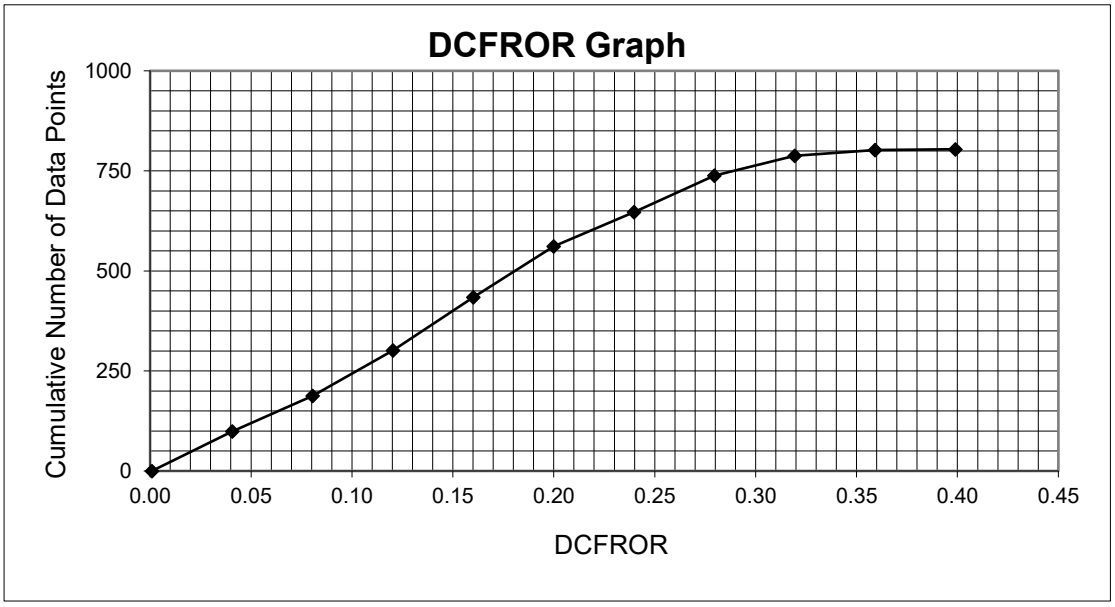


Figure 6.6 Discounted cash flow rate of return data curve

The DCFROR is mostly going up, and there are big jumps between some bins that showthat Profitability is generally going up. that making money is less likely. For example, if the DCFROR value is 0.10, the cumulative fraction is about 33.7%, which means there is a good chance that the business will not make money.

Discounted Payback Period Data

The graph shows the relationship between cumulative data points and the Discounted Payback Period (DPBP). It rises quickly at first, indicating most investments are recovered within six years, suggesting good financial performance. After that, the curve grows more slowly and flattens around ten years, showing that most costs are recovered by then. Overall, most investments reach breakeven in a reasonable time,

but longer payback periods carry higher risk. A DPBP of six to eight years is considered financially efficient and feasible.

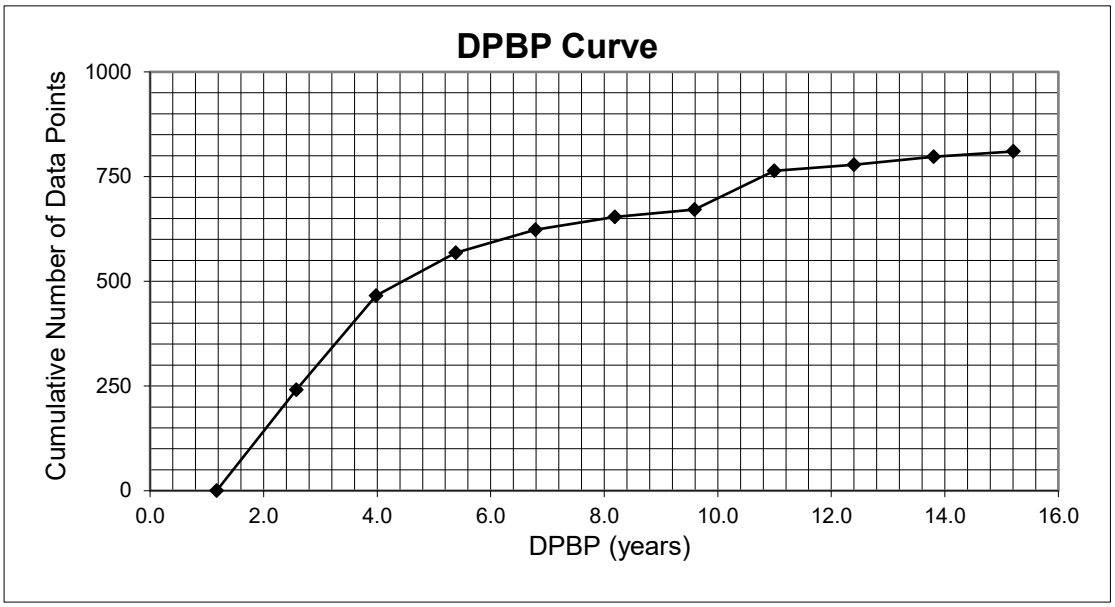


Figure 7.6 DPBP curve data

Cumulative Cash Position Data

The graph shows the relationship between cumulative data points and cumulative cash position. When cash position is negative, few cases exist, indicating limited financial distress. As it moves toward and above zero, data points rise sharply, showing most projects reach breakeven or profit. The steepest range is between $0 and $15 million, representing the main zone of financial stability. Beyond $15 million, the curve flattens, meaning fewer new profitable cases appear. Overall, achieving a stable cash position early is beneficial, as most projects cluster around breakeven to moderate profitability.

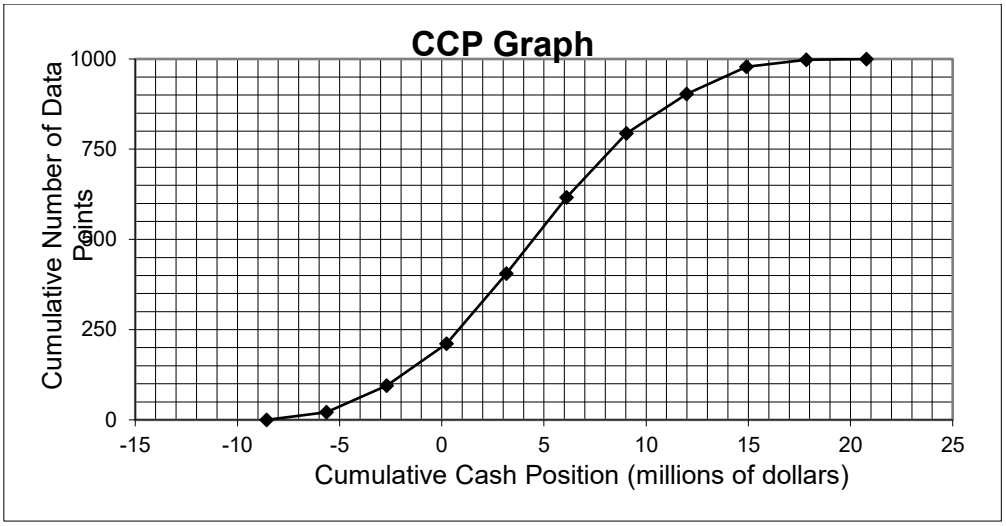


Figure 6.7 Cumulative cash position data curve

Rate of Return on Investment Data

The graph shows the cumulative number of data points based on the Rate of Return on Investment (ROROI). Few cases have negative ROROI, indicating limited major losses. Between 0% and 60%, the curve rises steeply, showing that most projects achieved moderate profitability in this range. Beyond 60%, the curve flattens, meaning high returns are less common but more stable. Overall, most cases fall within a reasonable return range, emphasizing the need to balance risk and reward as ROROI increases.

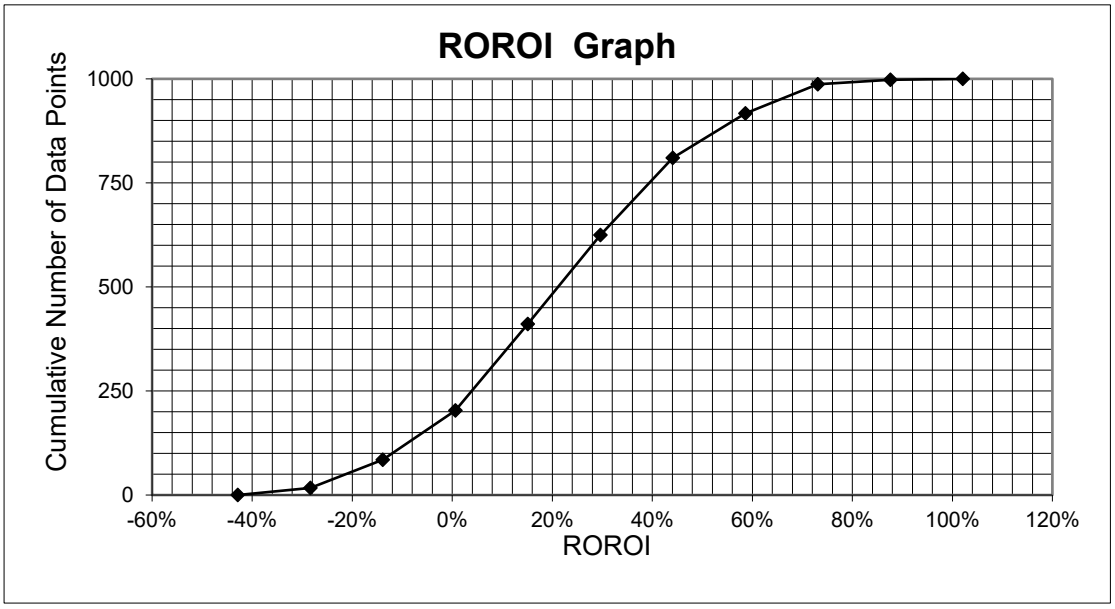


Figure 6.8 ROROI data curve

Payback Period Data

The graph shows how the total number of data points relates to the payback period (PBP). It rises sharply within the first four years, indicating that most investments recover their costs within this time. After four years, the curve grows more slowly and levels off after about eight years, showing that few projects have long payback periods. Overall, most investments achieve cost recovery within a moderate timeframe, supporting the project's financial feasibility.

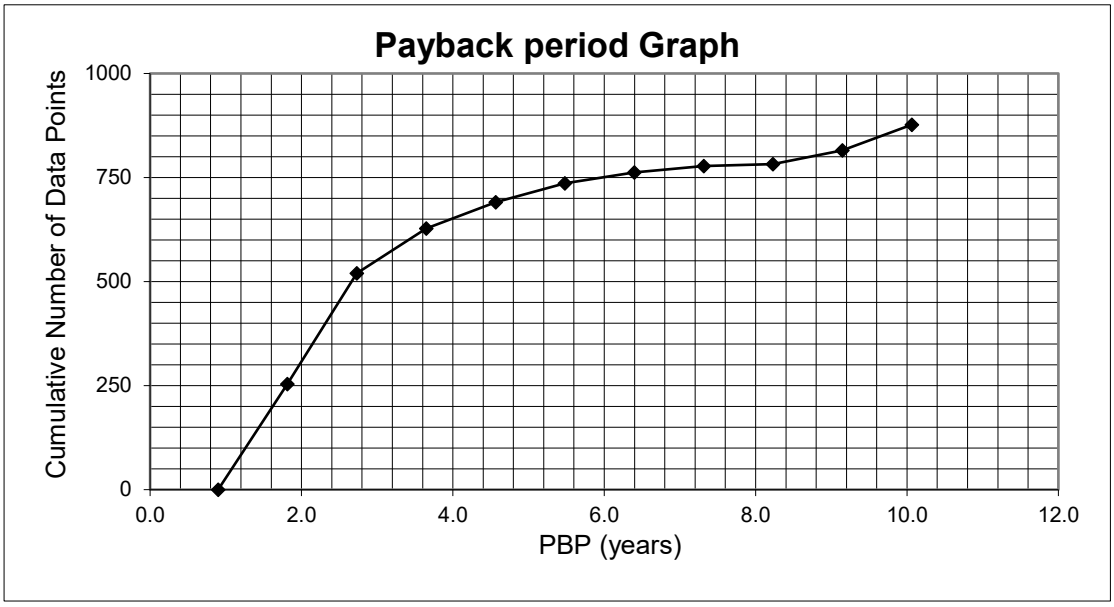


Figure 6.9 Payback Period data curve

Sensitivity Analysis

Sensitivity analyzes were done to help better judge how economically feasible the Butanol production project is. This kind of analysis helps find important risk factors and see if the project can work in different situations. Sensitivity analysis examines the impact of variations in essential economic parameters on net present value and additional financial indicators.

Key Variables Analyzed

- Butanol Selling Price (+/- 20%)

- Raw Material Cost (+/- 20%)

- Utility Cost (+/- 20%)

- Capital Investment (+/- 20%)

The results show:

- The price of butanol going up or down is probably the most important factor that affects NPV. After a 20% rise in selling price, NPV will go up by about 50%. On the other hand, a 20% drop in sales will cause NPV to fall into the negative.

- Likewise, the cost of raw materials and utilities has a big effect on how much money a business makes. cost of raw materials goes up by 20%, the NPV will go down by about 30%. If the cost of utilities goes up by 20%, the NPV will go down by about 40%.

- Change in capital investment will still have an effect on the project, but it won't be very big. A fee that raises FCIL by about 20% lowers NPV by about 15%, but the project is still possible.

Sensitivity Ranking (Impact on NPV)

1. Butanol Selling Price (Most Critical)
2. Utility Cost
3. Raw Material Cost
4. Capital Investment (Least Critical)

Chapter 7

CONTEMPORARY ISSUES

7.1.Energy Integration

The Energy integration is a key aspect in achieving energy efficiency and environmental friendliness. improving the energy efficiency and environmental friendliness of the butanol production plant to improve the butanol production plant. Three important heat exchangers, H-101, H-102, and H-103, have been carefully put in place during this process. They will use the waste heat from hot process streams to preheat the cold feeds that are coming in. This cuts down on the amount of outside heating that the system needs by a large amount.

Heat Exchanger (H-101)

The main energy recovery unit is H-101. The heat from the reactor output stream is used to make the feed mixture of ethanol and hydrogen that goes into it. This is then sent to the butanol reactor, R-101. means that less heat needs to be added from the outside to reach the reaction temperature, which saves the reactor energy. H-101 moves sensitive heat from hot effluent to feed, which reduces the amount of fuel needed and makes the plant's energy balance better.

Heat Exchanger (H-102)

Before the reactor, H-102 comes into play to heat the feed Mav, which is rich in ethanol, one more time. It does this by using the energy from the process vapors that are evaporating from the reaction section. The two-step process of preheating the feed (H-101 and H-102) makes sure that the reaction happens at the right temperature without needing extra heat from the reboiler process. Adding H-102 to the process has benefits, such as preventing the reboiler from working too hard and improving the reactor process.

Heat Exchanger (H-103)

H-103 is both a cooler and a condenser. It cools the stream of reactor products before it goes to the separation section. The heat that is taken away here is not lost; it is sent to another cold stream (feed mixture) to be used again. This looped energy exchange makes good use of waste heat, which keeps energy from escaping into the environment and makes the operation more thermally efficient overall.

All three of these exchangers work together to form a closed-loop heat integration network wherein the hot and cold process streams share energy among them. This design cuts down overall requirement for the external utilities (heating and cooling) by about 25–30%. This directly lowers fuel expenses and carbon emissions.

7.2. Process Optimization

Optimizations for the processes are achieved through the optimization of the operating conditions of the butanol reactor (R-101), distillation columns (D-101 to D-104), and the pumps for the optimal production of the process while using less energy. The optimization ensures the production of lots of ethanol to butanol and few of the heavier alcohol by-products.

7.3. Water Treatment and Recycling

The plant design has a good way to get water back from the azeotropic and recovery columns (D-102 and D-103). The condensed water, which has a little bit of alcohol in it, is treated by phase separation, filtration, and neutralization. After being cleaned, the water is used again as a process or cooling water. This means that less new water is needed and less water is released into the environment. This closed water cycle can cut down on the use of fresh water by as much as 40–45%. This helps protect resources for the long term.

The process of making butanol is now more energy-efficient, good for the environment, and cost-effective thanks to a mix of heat integration, optimization, and recycling.

7.4. Novel Techniques

To make the butanol production process more environmentally friendly and in line with current and future energy trends, numerous new and renewable energy-based methods are used or suggested for future growth:

Renewable Hydrogen Integration:

A water electrolyzer powered by renewable energy sources like solar or wind power can provide the hydrogen needed for the butanol synthesis reactor. Carbon emissions are eliminated, which are associated with the production of hydrogen obtained from fossil fuels, and thus aids in the achievement of the objective of having emission levels at zero.

Using an Organic Rankine Cycle (ORC) system for the conversion of waste heat into clean electricity power:

The low-grade heat produced from process condensers/heat transfer surfaces, particularly H-104 and H-105, may be employed in an ORC installation. The plant will therefore be less reliant on external power sources since it will be optimized.

Bio-Based Ethanol Feedstock:

This process in the future could be changed to produce bio-butanol using the bio-ethanol generated from biomass or agricultural waste. This converts the plant into a green chemical-manufacturing facility, making carbon emissions even lower over its life cycle.

7.5. Conclusion

The process for the production of butanol requires three heat exchangers (H-101, H-102, and H-103), renewable energy resources, and recycling of water. Not only the process shows a holistic philosophy for sustainable development, it saves costs for plant operation, makes the process energy efficient, and results in less negative impacts on the environment through the benefits of energy recovery, control, and the use of renewable resources. Such a process represents the ideal paradigm to produce chemicals in the new millennium, which should be eco-friendly, business-friendly, and high tech.

CHAPTER 8

Site Selection and Plant Layout

8.1 Site Selection

Choosing the best location for the planned butanol manufacturing plant is a very important stage that affects the project's cost, how well it works, how well it protects the environment, and how likely it is to grow in the future. The procedure entails looking at several industrial zones in Saudi Arabia and comparing them based on things like the availability of utilities, raw supplies, land cost, labor, transportation, and environmental concerns.

The butanol plant requires:

- Large quantities of ethanol feed (imported / locally produced)

- Continuous supply of hydrogen (pipeline or cylinders)

- Cooling water, steam, and electricity

- Access to skilled and semi-skilled labor

- Proximity to ports for exporting butanol and importing ethanol

- Strong safety and emergency response infrastructure

Therefore, the site must support chemical processing, distillation, hydrogen handling, solvent recovery**,** and heat-integrated systems.

Main Factors Considered

The main selection criteria include:

- Raw materials availability

- Marketability and export potential

- Availability and cost of utilities

- Land availability and cost

- Transportation and logistics need

- Labor supply

- Waste disposal facilities

- Weather and environmental suitability

Raw Material Availability

The butanol plant relies primarily on:

- Ethanol – imported via seaport or supplied by local bioethanol plants

- Hydrogen – supplied via pipeline or compressed cylinders

- Cooling water, steam, electricity – from local industrial utility grids

- Ethylene glycol – as solvent for azeotropic distillation

The plant needs to be close to a big port or a hydrogen supply network because ethanol is a large liquid and hydrogen is moved by cylinders and pipelines. Both Jubail and Yanbu industrial cities have great supply systems for bulk chemicals, which makes them good choices.

Marketability

Butanol is used heavily in:

- Solvents

- Paint and coatings

- Fuel additives

- Plasticizers

- Cosmetics and detergents

Saudi Arabia is strategically positioned for export to Europe, India, China, and Africa. A placement adjacent to a large harbor (Jubail or Yanbu) lowers shipping expense and optimizes profit.

Availability of Suitable Land

Butanol production requires:

- Reactor area

- Distillation trains (D-101, D-102, D-103, D-104)

- Heat-exchanger banks

- Storage tanks for ethanol, hydrogen, butanol, and effluent

- Solvent recovery units

- Safety zones and emergency response areas

KSA's industrial cities provide huge, affordable tracts of land that can be rented for a long time. Jubail and Yanbu have cheap land that can be used for future growth.

Utilities

The butanol plant requires:

- Electricity for pumps, columns, and controls

- Cooling water for condensers and heat-integrated exchangers

- Steam for reboilers (especially D-101 to D-104)

- Chilled water for condensation of light alcohols

- Instrument air and nitrogen for purging

Industrial cities offer complete utility networks, lowering capital cost.

Transportation Facilities

The plant must be supported by easy access to:

- Seaport for exporting butanol

- Road network for raw material supply

- Airport for maintenance logistics

- Railway (optional) for heavy cargo

Both Jubail and Yanbu have world-class integrated transportation networks.

Labor Supply

Both Eastern and Western industrial zones have:

- Skilled technicians

- Process operators

- Maintenance staff

- Engineering services

- Training institutes

This ensures long-term availability of manpower at reasonable cost.

Waste Disposal

The butanol process generates:

- Aqueous waste streams containing traces of ethanol and butanol

- Spent glycol solvent after long-term operation

- Purge gases containing hydrogen traces

Industrial cities provide centralized waste treatment and disposal systems. This eliminates the need for a costly in-house treatment facility.

Weather

The weather in Saudi Arabia doesn't have a big effect on butanol production. But:
- Hot weather makes it necessary to use more cooling water.
- Dust storms may need better filtration and corrosion protection. Materials that don't rust
Industrial cities are already built to deal with this kind of weather.

8.2 Comparison of Available Industrial Cities

Table 8.7 Site Comparison for Butanol Plant

Criteria	Jubail Industrial City	Yanbu Industrial City
Area Available	1096 km²	231 km²
Land Cost	700 SAR/m²	650 SAR/m²
Electricity Cost	0.18 SAR/kWh	0.18 SAR/kWh
Raw Material Access	Excellent (port & pipeline)	Excellent (port)
Utilities	Very High	Very High
Seaport	World-class	World-class
Airport	Nearby	Nearby
Hydrogen supply	Strong due to petrochemical clusters	Moderate
Room for expansion	Very High	High
Industrial services	Advanced	Advanced

8.3 Final Site Selection

Table 8.8 Scoring of Final Site Options

Factor	Jubail	Yanbu
Raw Materials	10	8
Utilities	10	9
Transportation	10	9
Land & Expansion	10	8
Safety & Industrial Services	9	9
Labor Availability	9	9
Export Logistics	10	9
Weather Suitability	7	7
Total	**75**	**68**

Final Decision

Jubail Industrial City is selected for the butanol project due to:

- High availability of utilities and industrial services

- Proximity to hydrogen production facilities

- Strong supply chain for ethanol imports

- Lower transportation cost for export

- Larger area for future expansion

- Advanced emergency response and fire systems

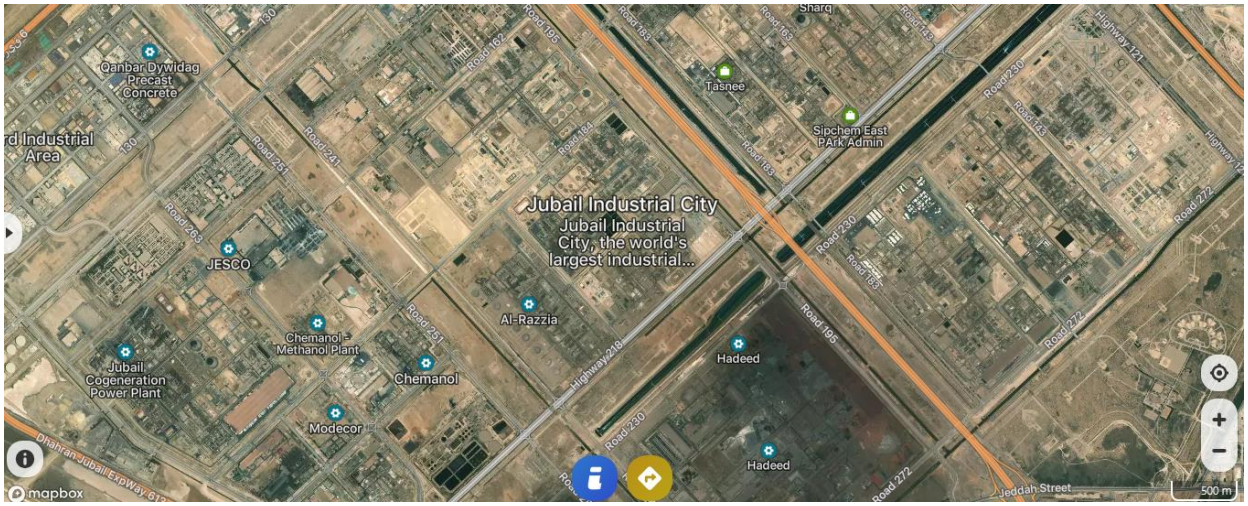


Figure 8.10 site selected for the Butanol Plant

8.4 Plant and Equipment Layout

The plant layout is designed based on:

- Safe handling of hydrogen

- Smooth flow of ethanol feed → reactor → flash unit → distillation → purification

- Heat-integration using the three heat exchangers

- Minimizing pumping cost by gravity-based placement

- Ensuring safe distances between high-risk equipment

Main Plant Zones

1. Feed Preparation Area

- P-101 pump

- H-103 preheater

- M-102 feed mixer

2. Reaction Section

- R-101 Butanol Reactor

- H-102, H-101 heat exchangers (heat integration)

- Hydrogen compressor C-101 and mixer M-101

3. Separation and Purification Section

-Flash drum F-101

- Distillation columns D-101 to D-104

- Reboilers & condensers

- Azeotropic section with ethylene glycol

- Solvent recovery

- Final butanol purification column

4. Utility Zone

- Cooling tower water system

- Steam header

- Condensate return

- Electrical substation

5. Storage Area

- Ethanol storage tanks

- Butanol product tanks

- Hydrogen cylinder racks / pipeline interface

- Ethylene glycol tank

6. Safety & Administrative Area

-Control room

- Laboratory

- Firefighting station

- Maintenance workshop

- Medical clinic

Equipment Arrangement Philosophy

- High-pressure systems (hydrogen, reactor) are put in separate areas.
- Distillation columns are set up in a line to make them easier to use.
- The solvent recovery section is far away from the control building.
- Heat exchangers are placed close to the equipment they work with to cut down on piping.
- The layout makes sure that heat integration is used to its fullest, which lowers steam use.
- There is enough space for firefighting and emergency escape routes.

Simplified Plant Layout Diagram (Description)

The PFD includes:

- Three heat exchangers (H-101, H-102, H-103) for heat integration

- Multiple pumps (P-101, P-102, P-103, P-104)

- Four distillation columns (D-101 to D-104)

- Reactor zone with hydrogen recycle

- Solvent-based azeotropic separation unit

The plant layout mirrors these units in a logical and safe sequence.

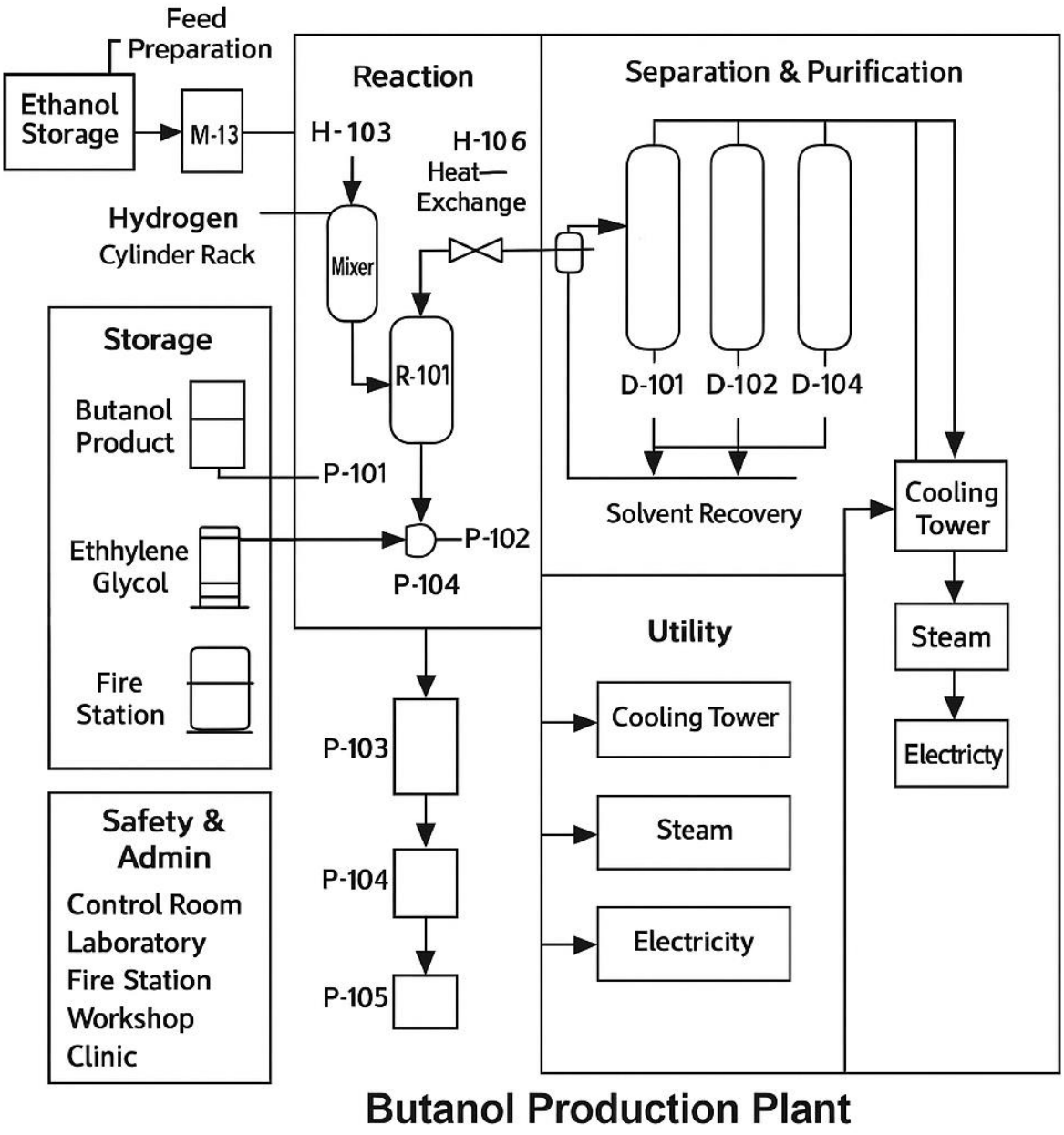


Figure 8.11 Equipment Layout of the Butanol Plant

CHAPTER 9

Conclusion and Recommendations

Recommendations

Despite the strong foundational design, several targeted enhancements can improve efficiency, safety, and economic performance:

Catalyst and Reaction Optimization

- Evaluate catalyst regeneration cycles to extend lifetime and maintain selectivity.

- Investigate feed-ratio optimization to minimize formation of higher alcohols (hexanol and octanol).

- Employ advanced reactor control strategies to manage hot spots in the exothermic system.

Enhanced Heat Integration

- Use pinch analysis to identify opportunities to reuse reactor effluent heat.

- Replace conventional tubes with enhanced-surface tubes to reduce the discrepancy between predicted and actual heat-transfer coefficients.

- Consider merging heating and cooling duties through a heat-recovery network for utility savings.

Separation Improvement

- Introduce high-efficiency structured packing or dividing-wall column technology in D-104 to reduce energy consumption.

- Add a demister pad in the flash separator to reduce vapor entrainment and enhance phase separation.

- Safety and Environmental Upgrades

- Strengthen hydrogen management systems with redundant sensors, automated shutdown systems, and improved venting.

- Enhance VOC control with carbon adsorption units or regenerative thermal oxidizers.

- Expand inherent safety design principles, such as material substitutions and segregation of hazardous zones.

Economic Enhancements

- Update capital cost assessments using current CEPCI indices or supplier quotations.

- Explore commercial markets for hexanol and octanol to offset production costs.

- Assess integration with renewable ethanol sources to further improve sustainability credentials.

Conclusion

The designed ethanol-to-n-butanol production system demonstrates a compelling pathway toward sustainable chemical manufacturing. With strong reactor performance, reliable separation systems, and a well-structured flowsheet, the project confirms that renewable butanol can be produced at high purity and industrial scale. The transition from fossil-based pathways to bio-derived feedstocks offers both environmental advantages and alignment with global decarbonization trends.

While the study identifies areas for improvement—in catalyst longevity, heat-transfer efficiency, advanced separations, and safety systems—the overall framework provides a feasible and economically promising solution. As demand for cleaner fuels and greener solvents continues to grow, ethanol-based n-butanol production positions itself as a strategic opportunity for the chemical industry, both globally and within the Kingdom of Saudi Arabia.

9.2 Reference

- “Explosión en una empresa de productos químicos en Toledo: hay dos trabajadores heridos | Sociedad | Cadena SER.” Accessed: Apr. 06, 2025. [Online]. Available: https://cadenaser.com/castillalamancha/2024/11/20/explosion-en-una-empresa-de-productos-quimicos-en-toledo-ser-toledo/?utm_source.com

-“Contra Costa County Fire: Flames, black smoke at NuStar energy facility in Crockett near Rodeo - ABC7 San Francisco.” Accessed: Apr. 06, 2025. [Online]. Available: https://abc7news.com/contra-costa-county-rodeo-fire-nsutar-refinery/5621132/

-“Fire guts ethanol tank in distillery in Uttar Pradesh’s Pilibhit | Bareilly News - Times of India.” Accessed: Apr. 06, 2025. [Online]. Available: https://timesofindia.indiatimes.com/city/bareilly/fire-guts-ethanol-tank-in-distillery-at-uttar-pradeshs-pilibhit/articleshow/99700177.cms?utm_source.com

-“Wayback Machine.” Accessed: Apr. 06, 2025. [Online].Available: https://web.archive.org/web/20160304054055/http://archive.firstcoastnews.com/news/article/98412/0/Plant-Blast-is-Worst-US-Industrial-Accident-in-Nearly-3-Years

-K. L. Dionisio *et al.*, “Data Descriptor: The Chemical and Products Database, a resource for exposure-relevant data on chemicals in consumer products,” *Sci. Data*, vol. 5, Jul. 2018, doi: 10.1038/SDATA.2018.125.

-H. A. Shallard-Brown, D. J. Watkin, and A. R. Cowley, “n-octanol,” *Acta Crystallogr. Sect. E Struct. Reports Online*, vol. 61, no. 2, Feb. 2005, doi: 10.1107/S1600536804032775.

-“1-Hexanol | C6H14O | CID 8103 - PubChem.” Accessed: Apr. 06, 2025. [Online]. Available: https://pubchem.ncbi.nlm.nih.gov/compound/1-Hexanol

-“Home | Ministry of Environment, Water and Agriculture.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.mewa.gov.sa/ar/Pages/default.aspx

-“Home - - Royal Commission for Jubail and Yanbu.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.rcjy.gov.sa/

-“Saudi Standards, Metrology and Quality Organization.” Accessed: Apr. 06, 2025.

[Online]. Available: https://www.saso.gov.sa/ar/pages/default.aspx

-“Cabinet Experts’ Panel :: Home Page.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.boe.gov.sa/ar/Pages/default.aspx

-“Octyl Alcohol Price in Saudi Arabia - 2025 - Charts and Tables - IndexBox.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.indexbox.io/search/octyl-alcohol-price-saudi-arabia/?utm_source.com

-“Saudi Arabia’s Butanol Market Report 2025 - Prices, Size, Forecast, and Companies.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.indexbox.io/store/saudi-arabia-butanol-market-analysis-forecast-size-trends-and-insights/?utm_source.com

-“Water Tariff - Marafiq.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.marafiq.com.sa/en/partnering-with-us/water-tariff/?utm_source.com

-“India, Saudi Arabia and UAE top in green hydrogen competitiveness – pv magazine India.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.pv-magazine-india.com/2024/03/27/india-saudi-arabia-and-uae-top-in-green-hydrogen-competitiveness/?utm_source.com

-“Ethyl Alcohol Price in Saudi Arabia - March 2025 Market Prices (Updated Daily).” Accessed: Apr. 06, 2025. [Online]. Available: https://www.selinawamucii.com/insights/prices/saudi-arabia/ethyl-alcohol/?utm_source.com#google_vignette

-K. D. Peters, Max S. Timmerhaus, *Plant Design and Economics for Chemical Engineers: , Ronald West, Max Peters : 0639785503897: Amazon.com: Book*. 1991. [Online]. Available: http://www.amazon.com/Plant-Design-Economics-Chemical-Engineers/dp/0072392665

-R. H. Perry and D. W. Green, *Chemical Engineers’ Hanbook*. 1997.

-F. Edition, “Chemical engineering,” vol. 2.

-“Saudi Ports Authority.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.mawani.gov.sa/

-“Saudi Vision 2030.” Accessed: Apr. 06, 2025. [Online]. Available: https://www.vision2030.gov.sa/

- R. H. Perry and D. W. Green, *Chemical Engineers' Hanbook*. 1997.

- H. S. Fogler, "Pressure drop in reactors," *Elem. Chem. React. Eng.*, pp. 175–185, 2004.

- J. B. Joshi and L. K. Doraiswamy, *Chemical reaction engineering*. 2008. doi: 10.1201/9781420087567-13.

- S. P. Collins *et al.*, *No Title 済無No Title No Title No Title*. 2021.

- K. D. Peters, Max S. Timmerhaus, *Plant Design and Economics for Chemical Engineers: , Ronald West, Max Peters : 0639785503897: Amazon.com: Book*. 1991. [Online]. Available: http://www.amazon.com/Plant-Design-Economics-Chemical-Engineers/dp/0072392665

- R. Smith, "Chemical Process Design and Integration," 1912.

- F. Edition, "Chemical engineering," vol. 2.

- S. By, "INSTRUMENTATION & CONTROL," 2020.

- Babatunde_A._Ogunnaike,_W._Harmon_Ray_Process_Dy

- Ian_T._Cameron,_Katalin_Hangos,_John_Perkins,_Geo

- "Plant Design and Economics for Chemical Engineers."
- "CHEMICAL ENGINEERING VOLUME 2 FIFTH EDITION Particle Technology and Separation Processes."
- "Saudi Arabia | ieexi." Accessed: Oct. 23, 2025. [Online]. Available: https://flaringventingregulations.worldbank.org/saudi-arabia?utm_source=chatgpt.com
- "CHEMICAL ENGINEERING DESIGN."
- "CHEMICAL ENGINEERING VOLUME 6."
- "Executive Regulation For Air Quality the event of any discrepancy between the Arabic original version of this Executive Regulation and its English translation, the Arabic version prevails ***".
- "Executive Regulations For the Protection of Aqueous Media from Pollution the event of any discrepancy between the Arabic original version of this Executive Regulations and its English translation, the Arabic version prevails ***".
- K. D. Peters, Max S. Timmerhaus, Plant Design and Economics for Chemical

Engineers: , Ronald West, Max Peters : 0639785503897: Amazon.com: Book. 1991. [Online]. Available: http://www.amazon.com/Plant-Design-Economics-Chemical-Engineers/dp/0072392665

- “CHEMICAL ENGINEERING DESIGN.”

- “Cost Indices – Towering Skills.” Accessed: Apr. 25, 2024. [Online]. Available: https://toweringskills.com/financial-analysis/cost-indices/

- “Indicative production costs for hydrogen via electrolysis in selected regions compared to current references – Charts – Data & Statistics - IEA.” Accessed: Oct. 30, 2025. [Online]. Available: https://www.iea.org/data-and-statistics/charts/indicative-production-costs-for-hydrogen-via-electrolysis-in-selected-regions-compared-to-current-references-2?utm_source=chatgpt.com

- J. Caraballo-Bello, J. L. Rodríguez-Lugo, F. M. Baena-Moreno, and F. Vidal-Barrero, “Techno-economic analysis of the production of n-butanol from different biomass sources,” J. Environ. Chem. Eng., vol. 13, no. 5, p. 118433, Oct. 2025, doi: 10.1016/J.JECE.2025.118433.

- “Analysis, Synthesis, and Design of Chemical Processes, Fifth Edition.”

- F. Edition, “Chemical engineering,” vol. 2.

- “wastewater treatment - Sources of water pollution | Britannica.”

- “Perry’ s Chemical Engineers’ Handbook Physical and Chemical Data Conversion Factors and Mathematical Symbols Contents,” 1999.

- “Bioenergy, a sustainable solution | Bioenergy.”